國立東華大學應用數學系

統計碩士班

碩士論文

時空 skew-t 過程於門檻超標預測之延伸：AR(2) 時間先驗與多尺度樣條基底函數

Extending the Spatio-Temporal Skew-t Process for Threshold Exceedance Prediction: AR(2) Temporal Priors and Multi-Resolution Spline Basis Functions

研究生：謝易軒

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致謝辭

完成這篇論文的一路上，受到許多人的幫助。感謝我的父母給予的信任及栽培，讓我可以專注地完成學業。謝謝黃灝勻老師在研究上給予的諸多方向及指點，每當我陷入瓶頸時，總能點出問題的關鍵，讓我學會如何思考與面對研究中的未知。謝謝吳韋瑩老師，無論是在運籌計畫或羽球場上，都教會了我面對挑戰的心態。謝謝蕭維政老師細心審閱論文，在口試時提供許多寶貴的建議，使這篇論文更加完整。也謝謝 302、314 研究室的夥伴們，在這段日子裡一起討論、互相打氣，讓漫長的研究過程多了許多歡笑；謝謝一路上陪伴我的朋友，在我感到疲憊時給予傾聽與鼓勵。

雖然很開心完成了這段碩士旅程，但在提筆致謝辭時卻感到點點不捨。這兩年所學到的，不只是專業知識，更是面對問題時的耐心與韌性。謝謝所有曾經幫助過我的人，我會帶著這份溫暖，繼續走向下一段旅程。

謝易軒 謹致

國立東華大學應用數學系統計碩士班

中華民國 115 年 7 月 25 日

摘要

在環境風險治理中，決策關心的通常不是平均濃度，而是未設站位置超越高門檻的機率。本文以 Morris et al. (2017) 的時空 skew-t 過程 (STP) 爲基準，沿兩軸檢驗其可改進空間，一是潛在參數的標準化 AR(2) 先驗，二是迴歸均值中的多尺度薄板樣條 (MRTS) 基底。模型比較以樣本外 Brier 分數爲核心指標，於模擬與 2005 年 7 月美國 AQS 臭氧資料（CMAQ 共變數）上進行。時間軸的結果依預測任務而分。空間內插對自迴歸階數不敏感；唯有當預測需外推至觀測紀錄之外，且生成機制的記憶延伸超過預測期距、其形式又能爲 AR(2) 先驗所刻畫時，第二階落後項才有回報。此增益在 $(\phi_1, \phi_2) = (0.15, 0.80)$ 下明確，而在自相關於預測期距內衰減與雙曲式衰減兩種情形下皆不存在。MRTS 均值項在非定態出現於均值曲面而非二階動差、且基底細緻到足以解析它時可改善預測；於臭氧資料上其名列前茅，惟未與其餘領先模型拉開差距。因此本文的貢獻不在於取代 STP 基準，而在於界定這兩項擴展各自在何種預測任務與何種非定態下值得採用，並以樣本外超標預測表現作爲取捨的依據。

關鍵字：時空 skew-t 過程、門檻超標預測、AR(2) 先驗、多尺度薄板樣條、Brier 分數、臭氧

Abstract

In environmental risk regulation, the inferential target is often not the field mean but the probability of exceeding a high threshold at unmonitored locations. We take the spatio-temporal skew-t process (STP) of Morris et al. (2017) as a baseline and examine where predictive improvement remains possible along two axes, a standardized AR(2) prior on the latent parameters and multi-resolution thin-plate spline (MRTS) basis functions in the regression mean. Models are compared by held-out Brier scores in simulation and on July 2005 U.S. AQS ozone data with CMAQ covariates. The temporal axis divides along the prediction task. Prediction at withheld sites is insensitive to the autoregressive order, whereas forecasting days beyond the end of the record rewards the second lag when the generating memory reaches past the horizon in a form the AR(2) prior can represent. The gain is decisive at $(\phi_1, \phi_2) = (0.15, 0.80)$ and absent both when the memory decays inside the horizon and under hyperbolic decay. The MRTS mean improves prediction where the non-stationarity lies in the mean surface rather than the second moment and the basis is fine enough to resolve it; on the ozone data it ranks among the leading models without separating from them. The contribution is therefore to enlarge the STP baseline into a candidate model class selected by held-out exceedance predictive performance.

Keywords: spatio-temporal skew-t process; threshold exceedance prediction; AR(2) prior; multi-resolution thin-plate splines; Brier score; ozone

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1 Introduction

Demand for accurate pollution-concentration estimates over broad spatial domains continues to grow, driven by epidemiological evidence tying air quality to morbidity and mortality and by tightening regulation, yet the cost and logistics of a monitoring network cap the coverage that direct measurement can provide. This gap motivates statistical methods that extend a network's spatial reach with quantified confidence (U.S. Environmental Protection Agency, 2024). Ground-level ozone is the canonical regulatory case. The National Ambient Air Quality Standard (NAAQS) specifies the maximum concentration of ozone (O_3) allowed in outdoor air. A site complies only if the three-year average of its annual fourth-highest daily maximum eight-hour concentration stays at or below the current NAAQS of 0.070 parts per million (70 ppb), last revised in 2015. The Clean Air Act requires the EPA to review the standard at least every five years; Table 1 lists its successive revisions. The policy-relevant target is thus not the mean field but its upper tail, namely the probability that an unmonitored location breaches a high threshold; our goal is to predict that probability at unmonitored sites and to map it across the domain.

Table 1: Ozone National Ambient Air Quality Standard

Form of the Standard (parts per million, ppm)	**1997**	**2008**	**2015**
Annual 4$^{\text{th}}$ highest daily max 8-hour average, averaged over three years	0.080	0.075	0.070

A spatial model for ozone threshold exceedances warrants special consideration, since standard spatial methods are likely to perform poorly. Two features of the problem are decisive. First, likelihood-based spatial modeling typically assumes a Gaussian process, which is appropriate when interest lies in mean behavior. The Gaussian distribution is light-tailed and symmetric. Daily ozone maxima are neither. Their residuals are heavy-tailed and are well described by a skew-t distribution (Morris, Reich, Emeric Thibaud, and Cooley, 2017). Second, our interest is in the dependence structure at high levels, which a covariance designed for mean behavior does not capture. A Gaussian process is asymptotically independent outside the degenerate case of perfect dependence, so the probability that two sites are jointly extreme decays to zero as the level increases. Nearby ozone monitors, however, tend to record high values on the same days, and this agreement weakens with distance (Morris, Reich, Emeric Thibaud, and Cooley, 2017). These considerations call for a heavy-tailed, asymmetric margin that retains asymptotic dependence. The skew-t distribution provides both. It is a subasymptotic model, capturing the joint tail decay at high but finite levels rather than the limiting dependence of extreme events (Raphaël Huser and Wadsworth, 2022). This is the setting the ozone data occupy.

Our approach also departs from threshold models based on extreme-value distributions, a well-developed part of the field in which extreme events are defined as exceedances over a high threshold. Anthony C. Davison and Smith (1990) model univariate exceedances with the generalized Pareto distribution. Ledford and J. A. Tawn (1996) address the multivariate case through a censored likelihood that distinguishes the ways in which a threshold may be exceeded. Wadsworth and J. A. Tawn (2012) and E. Thibaud, Mutzner, and A. C. Davison (2013) carried these ideas to spatial extremes, fitting models by a censored pairwise likelihood (Padoan, Ribatet, and Sisson, 2010). R. Huser and A. C. Davison (2014) extended them to space-time data. Such methods are grounded in extreme-value theory and presume a threshold high enough for extremal models to be valid and for extrapolation beyond the observed range. They are also computationally demanding and, in practice, restricted to small datasets. Neither feature suits the present application, since the 75 ppb threshold corresponds to roughly the 0.92 sample quantile of the daily ozone maxima (Morris, Reich, Emeric Thibaud, and Cooley, 2017), well short of the far tail that extremal models require, and the network, at $1{,}089$ monitoring stations, is far larger than such methods can handle. A complementary strategy retains the max-stable process while reducing its computational cost. Fully Bayesian inference for max-stable processes is cumbersome at scale. Morris, Reich, and Emeric Thibaud (2019) address this by truncating the spectral representation to a small set of empirical basis functions that carry the spatial dependence. Their construction renders inference feasible for large datasets and, as a by-product, dispenses with the assumption of stationarity.

In this thesis we build on the spatial skew-t process of Morris, Reich, Emeric Thibaud, and Cooley (2017). This process is not max-stable, yet it retains extremal dependence. Their model has four components. A skew-t margin, built by location-scale mixing of Gaussian processes, supplies heavy tails and asymmetry. A random spatial partition, redrawn each day, confines asymptotic dependence to nearby sites. A censored likelihood focuses inference on exceedances above a threshold. A transformed first-order autoregressive process carries dependence through time. We extend this baseline along two axes. On the temporal axis, the autoregressive prior links each day only to the one before, and we add a further lag to capture longer-range dependence in the record. On the spatial axis, the baseline assumes a stationary Matérn covariance, restrictive over a continental network. We therefore introduce the multi-resolution thin-plate spline basis of Tzeng and Huang (2018) as fixed-effect covariates in the mean, following the basis-function idea reviewed above, so as to relax stationarity at little additional cost. Each axis is a candidate extension rather than a replacement of the baseline, and together they are intended to improve prediction of threshold exceedances.

The remainder of the thesis is organized as follows. Section 2 reviews the spatial skew-t process of Morris, Reich, Emeric Thibaud, and Cooley (2017), develops the two extensions in turn (the standardized AR(2) prior on the latent dynamics and the MRTS basis functions in the mean), and assembles them into a single hierarchical model together with the MCMC algorithm used for inference. Section 3 evaluates each extension on synthetic data, describing the shared simulation design and the validation protocols, with the uncertainty attached to each score comparison, and reporting the AR(2) and MRTS experiments in turn. Section 4 applies the models to the ozone monitoring data, comparing the baseline against the two extensions in their ability to predict threshold exceedances. Section 5 concludes and outlines directions for further work. Technical derivations and design details are collected in the appendix.

2 Methodology

This section reviews the spatio-temporal skew-t process (STP) of Morris, Reich, Emeric Thibaud, and Cooley (2017), develops the AR(2) prior on the Gaussian-copula margins of its latent processes and the multi-resolution thin-plate spline (MRTS) augmentation of its regression mean, and assembles the full hierarchical model with the MCMC algorithm used for inference.

2.1 The spatio-temporal skew-t model

Let $Y_t(\mathbf{s})$ denote the response at spatial location $\mathbf{s} \in \mathcal{D} \subset \mathcal{R}^2$ on day $t \in \{1, \ldots, n_t\}$. Following Morris, Reich, Emeric Thibaud, and Cooley (2017), we model

$$Y_t(\mathbf{s}) \;=\; \mathbf{X}_t(\mathbf{s})^\top \boldsymbol{\beta} \;+\; \lambda\, \sigma_t(\mathbf{s})\, |z_t(\mathbf{s})| \;+\; \sigma_t(\mathbf{s})\, v_t(\mathbf{s}), \tag{1}$$

where $\mathbf{X}_t(\mathbf{s})$ is a p-vector of covariates, $\boldsymbol{\beta}$ is the corresponding regression coefficient vector, $\lambda \in \mathcal{R}$ is a scalar skewness parameter, and $v_t(\mathbf{s})$ is a standard Gaussian process with stationary isotropic Matérn correlation

$$\rho(h) \;=\; (1-\eta)\, I(h=0) \;+\; \eta\, \frac{1}{\Gamma(\nu)\, 2^{\nu-1}} \left(\sqrt{2\nu}\, \frac{h}{\varphi}\right)^{\nu} K_\nu\!\left(\sqrt{2\nu}\, \frac{h}{\varphi}\right), \tag{2}$$

where $\varphi > 0$ is the spatial range, $\nu > 0$ is the smoothness, $\eta \in [0, 1]$ is the proportion of variance attributed to the spatial component, K_ν is the modified Bessel function of the second kind, and $h = \|\mathbf{s}_1 - \mathbf{s}_2\|$ is the Euclidean distance between sites $\mathbf{s}_1$ and $\mathbf{s}_2$. Marginalizing over $|z_t(\mathbf{s})|$ and $\sigma_t(\mathbf{s})$, the finite-dimensional vector $\mathbf{Y}_t = [Y_t(\mathbf{s}_1), \ldots, Y_t(\mathbf{s}_n)]^\top$ follows an n-dimensional skew-t distribution; consequently (1) provides asymmetric and heavy-tailed margins together with asymptotic dependence between nearby sites.

Censoring. Because the EPA compliance criterion focuses on exceedances of a high level L, the model fits only the upper tail by censoring the data at a pre-specified threshold $T \leq L$. Writing the partially censored response $\widetilde{Y}_t(\mathbf{s}) = \max\{Y_t(\mathbf{s}), T\}$, inference is based on $\widetilde{\mathbf{Y}}_t = [\widetilde{Y}_t(\mathbf{s}_1), \ldots, \widetilde{Y}_t(\mathbf{s}_n)]^\top$, and the censored values below T are imputed inside the MCMC.

Random Partition. To remove the long-range asymptotic dependence that arises when a single scalar z and σ^2 are shared across the entire domain, the model uses a daily random partition. For each day t, let $\mathbf{w}_{t1}, \ldots, \mathbf{w}_{tK}$ denote K spatial knots in $\mathcal{D}$, and define the partition subregions

$$P_{tk} = \{\mathbf{s} \in \mathcal{D} : k = \arg\min_\ell \|\mathbf{s} - \mathbf{w}_{t\ell}\|\}, \qquad k = 1, \ldots, K. \tag{3}$$

For $\mathbf{s} \in P_{tk}$ we set $z_t(\mathbf{s}) = z_{tk}$ and $\sigma_t^2(\mathbf{s}) = \sigma_{tk}^2$, with $z_{tk} \overset{iid}{\sim} N(0, 1)$ and $\sigma_{tk}^2 \overset{iid}{\sim} \mathrm{IG}(a/2, b/2)$. Within each subregion the model retains the skew-t form, while sites in different partitions are not asymptotically dependent.

Temporal Dependence. Daily temporal dependence is introduced on the latent parameters that drive the tail, namely the knot locations $\mathbf{w}_{tk}$, the skewness latents z_{tk}, and the scale parameters σ_{tk}^2. To preserve the marginal skew-t distribution at every day, each parameter is first transformed to a standard normal scale via the probability integral transform (PIT):

$$w_{tki}^* = \Phi^{-1}\left[\frac{w_{tki} - \min(\mathbf{s}_i)}{\max(\mathbf{s}_i) - \min(\mathbf{s}_i)}\right], \quad i = 1, 2, \tag{4}$$

$$z_{tk}^* = \Phi^{-1}\{\mathrm{HN}(z_{tk})\}, \tag{5}$$

$$\sigma_{tk}^{2*} = \Phi^{-1}\{\mathrm{IG}(\sigma_{tk}^2)\}, \tag{6}$$

where Φ is the standard normal distribution function, HN is the half-normal distribution function, and IG is the inverse-gamma distribution function with parameters $(a/2, b/2)$. After transformation, each $\mathbf{w}^*_{tk}, z^*_{tk}, \sigma^{2*}_{tk}$ has a standard normal marginal distribution, and Morris, Reich, Emeric Thibaud, and Cooley (2017) place a stationary AR(1) time series on each of them:

$$\mathbf{w}^*_{tk} \mid \mathbf{w}^*_{t-1,k} \sim N_2\big(\phi^{(w)}\, \mathbf{w}^*_{t-1,k},\ (1-(\phi^{(w)})^2)\, \mathbf{I}_2\big)\,, \tag{7}$$

$$z^*_{tk} \mid z^*_{t-1,k} \sim N\big(\phi^{(z)}\, z^*_{t-1,k},\ 1-(\phi^{(z)})^2\big)\,, \tag{8}$$

$$\sigma^{2*}_{tk} \mid \sigma^{2*}_{t-1,k} \sim N\big(\phi^{(\sigma)}\, \sigma^{2*}_{t-1,k},\ 1-(\phi^{(\sigma)})^2\big)\,, \tag{9}$$

with $|\phi^{(w)}|, |\phi^{(z)}|, |\phi^{(\sigma)}| < 1$, and initial values $\mathbf{w}^*_{1k} \sim N_2(\mathbf{0}, \mathbf{I}_2)$, $z^*_{1k} \sim N(0,1)$, $\sigma^{2*}_{1k} \sim N(0,1)$. Transforming back through (4)–(6) gives $\mathbf{w}_{tk} \sim \text{Unif}(\mathcal{D})$, $z_{tk} \sim \text{HN}(0,1)$, and $\sigma^2_{tk} \sim \text{IG}(a/2, b/2)$ at every t, so that the spatial skew-t structure of (1) is preserved at each day.

2.2 Extremal dependence

For exceedance prediction, what matters is how strongly the skew-t process couples extreme values at two sites, and how that coupling changes with their separation. One measure of this extremal dependence is the χ statistic (Coles, Heffernan, and J. Tawn, 1999), which for a stationary and isotropic spatial process at separation h is

$$\chi(h) = \lim_{c \to c^*} \Pr[Y(\mathbf{s}+h) > c \mid Y(\mathbf{s}) > c], \tag{10}$$

where c^* is the upper limit of the support of Y; for the skew-t distribution $c^* = \infty$. If $\chi(h) = 0$, then observations are asymptotically independent at distance h. For Gaussian processes, $\chi(h) = 0$ regardless of the distance h, so they are not suitable for modeling asymptotically dependent extremes. Unlike the Gaussian process, the skew-t process is asymptotically dependent. However, one problem with the spatial skew-t process is that $\lim_{h\to\infty} \chi(h) > 0$. This occurs because all observations, both near and far, share the same z and σ terms. Such long-range dependence is undesirable over large domains, where only local dependence is expected. Morris, Reich, Emeric Thibaud, and Cooley (2017) avoid this problem through a random partitioning mechanism, described in detail in Section 2.1.

2.3 Extension to AR(2) latent processes

In this section, we generalize the AR(1) prior (7)–(9) on the transformed latents to a stationary AR(2) prior.

The AR(2) prior. Writing $\{Y_t\}$ for any scalar component of $\mathbf{w}^*_{tk}$, or for z^*_{tk} or σ^{2*}_{tk} at a fixed knot k, the prior is the stationary AR(2) recursion

$$Y_t \;=\; \phi_1 Y_{t-1} \;+\; \phi_2 Y_{t-2} \;+\; \epsilon_t, \qquad \epsilon_t \overset{iid}{\sim} N(0, \sigma^2_\epsilon), \qquad t \geq 3, \tag{11}$$

with (ϕ_1, ϕ_2) confined to the stationarity triangle

$$\mathcal{S} \;=\; \big\{(\phi_1, \phi_2) \;:\; |\phi_2| < 1,\;\; \phi_1 + \phi_2 < 1,\;\; \phi_2 - \phi_1 < 1\big\}. \tag{12}$$

Because the inverse PIT (4)–(6) returns the half-normal, inverse-gamma, and uniform target margins only when $\gamma_0 := \mathrm{Var}(Y_t) = 1$, the innovation variance is not free. Solving the Yule–Walker equations under $\gamma_0 = 1$ (Appendix B) fixes the standardized autocovariances

$$\gamma_1 \;=\; \frac{\phi_1}{1 - \phi_2}, \qquad \gamma_2 \;=\; \phi_1 \gamma_1 + \phi_2, \tag{13}$$

$$\sigma^2_\epsilon \;=\; 1 - \phi_1 \gamma_1 - \phi_2 \gamma_2. \tag{14}$$

Setting $\phi_2 = 0$ gives $\gamma_1 = \phi_1$ and $\sigma^2_\epsilon = 1 - \phi_1^2$, the AR(1) form of (7)–(9), so the AR(2) prior strictly nests the prior of Morris, Reich, Emeric Thibaud, and Cooley (2017).

Stationary initial pair. The recursion (11) needs two seeds, and independent $N(0, 1)$ seeds would leave $\mathrm{Var}(Y_t) \neq 1$ over the first days and break the margin. We therefore draw the initial pair from the stationary bivariate Gaussian

$$\begin{pmatrix} Y_1 \\ Y_2 \end{pmatrix} \;\sim\; N_2\!\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 & \gamma_1 \\ \gamma_1 & 1 \end{pmatrix} \right), \tag{15}$$

so that $\mathrm{Var}(Y_t) = 1$ holds from $t = 1$ onward.

Application to the three transformed parameters. Mirroring the AR(1) recursions (7)–(9), the AR(2) prior (11) with initial condition (15) replaces it component-wise:

$$\mathbf{w}^*_{tk} \mid \mathbf{w}^*_{t-1,k}, \mathbf{w}^*_{t-2,k} \sim N_2\Big(\phi_1^{(w)}\, \mathbf{w}^*_{t-1,k} + \phi_2^{(w)}\, \mathbf{w}^*_{t-2,k},\ \sigma^2_{\epsilon,w}\, \mathbf{I}_2\Big)\,, \tag{16}$$

$$z^*_{tk} \mid z^*_{t-1,k}, z^*_{t-2,k} \sim N\Big(\phi_1^{(z)}\, z^*_{t-1,k} + \phi_2^{(z)}\, z^*_{t-2,k},\ \sigma^2_{\epsilon,z}\Big)\,, \tag{17}$$

$$\sigma^{2*}_{tk} \mid \sigma^{2*}_{t-1,k}, \sigma^{2*}_{t-2,k} \sim N\Big(\phi_1^{(\sigma)}\, \sigma^{2*}_{t-1,k} + \phi_2^{(\sigma)}\, \sigma^{2*}_{t-2,k},\ \sigma^2_{\epsilon,\sigma}\Big)\,, \tag{18}$$

where each pair $\boldsymbol{\phi}^{(\cdot)}$ lies in the stationarity triangle $\mathcal{S}$ of (12), the innovation variances follow from (13)–(14), and the initial pairs follow (15) component-wise. After the inverse PIT (4)–(6), the back-transformed $\mathbf{w}_{tk}, z_{tk}, \sigma^2_{tk}$ keep the same marginals as the AR(1) baseline, so the spatial skew-t model (1) is preserved at every t. The Metropolis step for each Y_t now targets a full conditional carrying three prior factors (self, lag-1, lag-2); its explicit form and the Metropolis–Hastings acceptance ratio are given in Appendix C.

2.4 Incorporating MRTS covariates

Tzeng and Huang (2018) proposed the multi-resolution thin-plate spline (MRTS) basis, whose functions are ordered from coarse to increasingly fine resolution. This ordering lets a model add spatial detail one resolution at a time; we define the basis below on a 2-dimensional spatial domain. Consider n distinct locations, $\mathbf{s}_1, \cdots, \mathbf{s}_n$, where $\mathbf{s}_i = (x_{i1}, x_{i2})^\top \in \mathcal{R}^2$ for $i = 1, \ldots, n$, and let

$$\mathbf{R} = \begin{pmatrix} 1 & x_{11} & x_{12} \\ \vdots & \vdots & \vdots \\ 1 & x_{n1} & x_{n2} \end{pmatrix}. \tag{19}$$

Then the MRTS basis functions $f_k(\mathbf{s})$ at a location $\mathbf{s} = (x_1, x_2)^\top$ are defined as

$$f_k(\mathbf{s}) = \begin{cases} 1, & k = 1, \\ x_{k-1}, & k = 2, 3, \\ \mu_{k-3}^{-1}\big\{\boldsymbol{\psi}(\mathbf{s}) - \boldsymbol{\Psi}\mathbf{R}(\mathbf{R}^\top\mathbf{R})^{-1}\,\mathbf{r}\big\}^\top \mathbf{v}_{k-3}, & k = 4, \ldots, n, \end{cases} \tag{20}$$

where $\mathbf{r} = (1, x_1, x_2)^\top$, $\boldsymbol{\psi}(\mathbf{s}) = (\psi_1(\mathbf{s}), \ldots, \psi_n(\mathbf{s}))^\top$ with

$$\psi_i(\mathbf{s}) \;=\; \frac{1}{8\pi}\,\|\mathbf{s}-\mathbf{s}_i\|^2\,\log\bigl(\|\mathbf{s}-\mathbf{s}_i\|\bigr), \qquad i = 1, \ldots, n, \tag{21}$$

and $\boldsymbol{\Psi}$ is the $n \times n$ matrix with (i, j)-th element $\psi_j(\mathbf{s}_i)$, $\mathbf{v}_k$ is the k-th column of $\mathbf{V}$, and $\mathbf{V}\,\mathrm{diag}(\mu_1, \ldots, \mu_n)\,\mathbf{V}^\top$ is the eigen-decomposition of $\mathbf{Q}\,\boldsymbol{\Psi}\,\mathbf{Q}$ with $\mu_1 \geq \cdots \geq \mu_n$ and $\mathbf{Q} = \mathbf{I} - \mathbf{R}(\mathbf{R}^\top\mathbf{R})^{-1}\mathbf{R}^\top$.

The MRTS basis enters the STP through the regression component. Fix the number of basis functions $K_{\text{MRTS}} \geq 1$ and let

$$\mathbf{f}(\mathbf{s}) \;=\; \bigl(f_1(\mathbf{s}),\, f_2(\mathbf{s}),\, \ldots,\, f_{K_{\text{MRTS}}}(\mathbf{s})\bigr)^\top \tag{22}$$

collect the first K_{MRTS} basis functions defined in (20). Then the day-t covariate vector $\mathbf{X}_t(\mathbf{s})$ in (1) is replaced by

$$\begin{aligned} \mathbf{X}_t^*(\mathbf{s}) &= \bigl[\mathbf{f}(\mathbf{s})^\top,\ \mathbf{X}_t(\mathbf{s})^\top\bigr]^\top \\ &= \bigl[f_1(\mathbf{s}),\ f_2(\mathbf{s}),\ \ldots,\ f_{K_{\text{MRTS}}}(\mathbf{s}),\ \mathrm{X}_1(\mathbf{s}, t),\ \ldots,\ \mathrm{X}_p(\mathbf{s}, t)\bigr]^\top, \end{aligned} \tag{23}$$

where $\mathrm{X}_j(\mathbf{s}, t)$ represents the j-th exogenous covariate of $\mathbf{X}_t(\mathbf{s})$. The model in (1) can then be expressed as

$$Y_t(\mathbf{s}) \;=\; \mathbf{X}_t^*(\mathbf{s})^\top\boldsymbol{\beta}^* \;+\; \lambda\,\sigma_t(\mathbf{s})\,|z_t(\mathbf{s})| \;+\; \sigma_t(\mathbf{s})\,v_t(\mathbf{s}), \tag{24}$$

which is identical to the STP model (1) on all components other than the mean, with $\boldsymbol{\beta}^* \in \mathcal{R}^{K_{\text{MRTS}}+p}$. Because the MRTS basis functions depend only on location, each site $\mathbf{s}$ shares the same ordered set of basis values $\mathbf{f}(\mathbf{s})$ at every day t. The number of basis functions K_{MRTS} is the practical tuning parameter. Section 3.4 compares $K_{\text{MRTS}} \in \{0, 3, 5, 8, 10, 12, 15, 20, 25\}$ across its simulation designs, extended by $\{30, 40, 50\}$ on the two nonlinear-mean designs.

2.5 Hierarchical model

We specify the model as a three-stage Bayesian hierarchical model in the sense of Berliner (1996), following the presentation of hierarchical spatial models in Gelfand et al. (2010), namely a data stage conditional on the latent process, a process stage conditional on the parameters, and a parameter stage carrying the priors. Collect the day-t latent processes as $\mathcal{Z}_t = \{(\mathbf{w}_{tk},\, z_{tk},\, \sigma^2_{tk}) : k = 1, \ldots, K\}$ and the parameters as $\boldsymbol{\theta} = \{\boldsymbol{\beta}, \lambda, \varphi, \nu, \eta, a, b, \boldsymbol{\phi}\}$, where $\boldsymbol{\phi}$ collects the AR coefficients of the three latent processes. The number of knots K is not treated as random; it indexes candidate models selected by held-out Brier score (Section 3.2).

Stage 1: data model. The observations are the censored responses,

$$\widetilde{Y}_t(\mathbf{s}_i) \;=\; \max\{Y_t(\mathbf{s}_i),\, T\}, \qquad i = 1, \ldots, n, \quad t = 1, \ldots, n_t, \tag{25}$$

so that values above T are observed exactly, while values below T are known only to lie below T and are imputed within the MCMC.

Stage 2: process model. The first substage is the conditional Gaussian field: independently across days,

$$\mathbf{Y}_t \mid \mathcal{Z}_t, \boldsymbol{\theta} \;\sim\; N_n\Big(\mathbf{X}_t\boldsymbol{\beta} + \lambda\, \mathbf{D}_t\, |\mathbf{z}_t|,\;\; \mathbf{D}_t\, \mathbf{R}(\varphi, \nu, \eta)\, \mathbf{D}_t\Big), \tag{26}$$

where $\mathbf{D}_t = \operatorname{diag}\{\sigma_t(\mathbf{s}_1), \ldots, \sigma_t(\mathbf{s}_n)\}$, $|\mathbf{z}_t| = (|z_t(\mathbf{s}_1)|, \ldots, |z_t(\mathbf{s}_n)|)^\top$, and $\mathbf{R}(\varphi, \nu, \eta)$ has entries $\rho(\|\mathbf{s}_i - \mathbf{s}_j\|)$ from (2). Under the MRTS extension, $\mathbf{X}_t\boldsymbol{\beta}$ is replaced by $\mathbf{X}_t^*\boldsymbol{\beta}^*$ of (23).

The second substage maps the subregion latents to the sites through the partition: for $\mathbf{s} \in P_{tk}$,

$$z_t(\mathbf{s}) \;=\; z_{tk}, \qquad \sigma_t^2(\mathbf{s}) \;=\; \sigma_{tk}^2, \qquad P_{tk} \;=\; \{\mathbf{s} \in \mathcal{D} : k = \arg\min_\ell \|\mathbf{s} - \mathbf{w}_{t\ell}\|\}. \tag{27}$$

The third substage gives the latent temporal dynamics on the transformed scale of (4)–(6), independently across knots $k = 1, \ldots, K$: for $t \geq 3$,

$$\mathbf{w}_{tk}^* \mid \mathbf{w}_{t-1,k}^*,\, \mathbf{w}_{t-2,k}^*,\, \boldsymbol{\phi}^{(w)} \sim N_2\big(\phi_1^{(w)}\mathbf{w}_{t-1,k}^* + \phi_2^{(w)}\mathbf{w}_{t-2,k}^*,\; \sigma_{\epsilon,w}^2\, \mathbf{I}_2\big), \tag{28}$$

$$z_{tk}^* \mid z_{t-1,k}^*,\, z_{t-2,k}^*,\, \boldsymbol{\phi}^{(z)} \sim N\big(\phi_1^{(z)} z_{t-1,k}^* + \phi_2^{(z)} z_{t-2,k}^*,\; \sigma_{\epsilon,z}^2\big), \tag{29}$$

$$\sigma_{tk}^{2*} \mid \sigma_{t-1,k}^{2*},\, \sigma_{t-2,k}^{2*},\, \boldsymbol{\phi}^{(\sigma)} \sim N\big(\phi_1^{(\sigma)}\sigma_{t-1,k}^{2*} + \phi_2^{(\sigma)}\sigma_{t-2,k}^{2*},\; \sigma_{\epsilon,\sigma}^2\big), \tag{30}$$

with each innovation variance fixed by the Yule–Walker standardization (13)–(14) and each initial pair drawn from the stationary bivariate Gaussian (15). The AR(1) baseline of Morris, Reich, Emeric Thibaud, and Cooley (2017) is the special case $\phi_2^{(\cdot)} = 0$, which reduces (28)–(30) to (7)–(9). The inverse transforms (4)–(6) return the marginals $\mathbf{w}_{tk} \sim \operatorname{Unif}(\mathcal{D})$, $z_{tk} \sim \operatorname{HN}(0, 1)$, and $\sigma_{tk}^2 \sim \operatorname{IG}(a/2,\, b/2)$ at every t.

Stage 3: parameter model. All parameters, and only parameters, receive priors at this stage:

$$\begin{aligned} &\boldsymbol{\beta} \sim N_p(\mathbf{0},\, \sigma_\beta^2 \mathbf{I}), && \lambda \sim N(0,\, \sigma_\lambda^2), && \eta \sim \text{Unif}(0, 1), \\ &\varphi \sim \text{Unif}(0,\, \varphi_{\text{max}}), && a/2 \sim \text{Unif}\{0.05,\, 0.10,\, \ldots,\, 5\}, && b/2 \sim \text{Gamma}(1,\, 1), \\ &\log \nu \sim N(-1.2,\, 1),\ \nu \le 10, && \boldsymbol{\phi}^{(\cdot)} \sim N_2\big(\mathbf{0},\, \sigma_\phi^2 \mathbf{I}_2\big)\, I\big(\boldsymbol{\phi}^{(\cdot)} \in \mathcal{S}\big). && \end{aligned} \tag{31}$$

Here Unif$\{\cdot\}$ denotes the discrete uniform distribution on the listed grid, and the hyperparameters are $\sigma_\beta = \sigma_\lambda = 20$ and $\sigma_\phi = 0.5$, with the range bound $\varphi_{\text{max}} = 15$ in the simulation study and $\varphi_{\text{max}} = 5$ in the ozone study, matched to the extent of each spatial domain. The smoothness ν receives a lognormal prior with median $e^{-1.2} \approx 0.30$, truncated above at 10. Both autoregressive coefficients carry the prior scale σ_ϕ of the AR(1) baseline, so the AR(1) and AR(2) fits are matched on prior scale rather than tuned separately. One set of fits departs from (31), namely the time-block forecast fits of Section 3.3, which truncate the prior on λ to the positive half line for an identifiability reason given there; every other fit reported in this thesis uses (31) as stated.

Writing $[\,\cdot\,]$ for a generic distribution, the three stages multiply to the joint model, and by Bayes' Rule the posterior is proportional to that product,

$$\big[\mathbf{Y}_{\text{cens}},\, \mathcal{Z},\, \boldsymbol{\theta} \mid \widetilde{\mathbf{Y}}\big] \;\propto\; \prod_{t=1}^{n_t} \big[\widetilde{\mathbf{Y}}_t \mid \mathbf{Y}_t\big]\, \big[\mathbf{Y}_t \mid \mathcal{Z}_t,\, \boldsymbol{\theta}\big]\; \big[\mathcal{Z}_{1:n_t} \mid \boldsymbol{\theta}\big]\, \big[\boldsymbol{\theta}\big], \tag{32}$$

sampled by a Gibbs scan that draws directly from every closed-form full conditional, namely those of $\boldsymbol{\beta}$, λ, $a/2$, $b/2$, and the censored responses. The remaining blocks are updated by Metropolis–Hastings steps, namely the latent processes on the transformed scale of (4)–(6), the autoregressive coefficients, and the Matérn parameters. A Gibbs scan of this kind, with Metropolis–Hastings steps standing in for the intractable draws, is a Metropolis-within-Gibbs sampler. Each factor of (32) is one of the stages above, and the full conditionals are read off stage by stage; the full conditionals of the time-varying parameters and of the autoregressive coefficients, together with their Metropolis–Hastings acceptance ratios, are given in Appendix C. The predictive distributions that this posterior induces are given in Section 2.6.

2.6 Posterior predictive distribution

The two validation protocols of Section 3.2 both score a posterior predictive distribution, and they differ in which of its factors the data can inform. Write $\widetilde{\mathbf{Y}}_{1:n_o}$ for the record observed up to time n_o, let $\mathbf{Y}^{\text{new}}$ collect the responses to be predicted, and let $\mathcal{Z}^{\text{new}}$ collect whatever latent processes those responses require beyond $\mathcal{Z}_{1:n_o}$. Integrating the joint model (32) over the posterior gives

$$\left[\mathbf{Y}^{\text{new}} \mid \widetilde{\mathbf{Y}}_{1:n_o}\right] = \int \underbrace{\left[\mathbf{Y}^{\text{new}} \mid \mathcal{Z}^{\text{new}},\, \mathcal{Z}_{1:n_o},\, \boldsymbol{\theta}\right]}_{\text{data model}} \underbrace{\left[\mathcal{Z}^{\text{new}} \mid \mathcal{Z}_{1:n_o},\, \boldsymbol{\theta}\right]}_{\text{latent transition}} \underbrace{\left[\mathcal{Z}_{1:n_o},\, \boldsymbol{\theta} \mid \widetilde{\mathbf{Y}}_{1:n_o}\right]}_{\text{posterior}} \, \mathrm{d}\mathcal{Z}^{\text{new}} \, \mathrm{d}\mathcal{Z}_{1:n_o} \, \mathrm{d}\boldsymbol{\theta}. \tag{33}$$

Only the third factor is a posterior. The second is the AR(2) prior (28)–(30) of the process stage, which the data reach only through ϕ and through the states it is conditioned on. Which of the two remaining factors carries the prediction is what separates the protocols.

Prediction at an unmonitored site. When the target is a day $t \leq n_o$ at a site $\mathbf{s}_0$ that was not monitored, $\mathcal{Z}^{\text{new}}$ is empty and the transition factor drops out of (33). Day t is observed at the monitored sites, so z_{tk} and σ^2_{tk} are already part of $\mathcal{Z}_{1:n_o}$ and carry a posterior of their own. Within a draw, $Y_t(\mathbf{s}_0)$ then follows the Gaussian conditional of (26) given the monitored responses of that day, whose mean adds to $\mathbf{X}_t(\mathbf{s}_0)^\top\boldsymbol{\beta} + \lambda\sigma_t(\mathbf{s}_0)|z_t(\mathbf{s}_0)|$ a term proportional to the day-t residuals at the monitored sites. This is Bayesian kriging, and the cross-sectional information of the day being predicted enters through that term.

Forecast beyond the end of the record. When the target is a day $n_o + h$, no site is observed on that day, the kriging term of the previous paragraph has nothing to condition on, and $\mathcal{Z}^{\text{new}} = \mathcal{Z}_{n_o+1:n_o+H}$ must be supplied entirely by the transition factor. The AR(2) recursion is therefore run forward from the two seam states $\mathcal{Z}_{n_o-1}$ and $\mathcal{Z}_{n_o}$ of the same draw, on the transformed scale of (4)–(6) where the recursion is defined, and the inverse transforms return the latents to the scale on which (26) acts. Because the autocorrelations (13) decay in h, the forecast distribution relaxes toward the stationary margins of (4)–(6) as the lead grows, and the persistence of the fitted process governs how quickly. A model whose latent processes are independent in time is the limiting case in which each lead is drawn from those margins directly, so under that model the forecast carries no information from the record beyond $\boldsymbol{\theta}$.

Algorithm 1 states the Monte Carlo form of the forecast. Each retained draw contributes one path, so parameter uncertainty and seam-state uncertainty are both propagated and no posterior summary is substituted for either. Conditioning on the seam states of the draw, rather than on the stationary distribution of the latent processes, is what makes the AR(1) and AR(2) forecasts differ from the independent one at short leads. The exceedance probability $\widehat{P}(Y > L)$ required by the Brier score of Section 3.2 is the proportion of the returned paths that exceed L at the site and lead being scored.

Algorithm 1 Posterior predictive forecast over the forecast window $(n_o, n_o + H]$.

1: **for** each retained posterior draw $m = 1, \ldots, M$ **do**
2: take $\boldsymbol{\theta}^{(m)}$ and the seam states $\mathcal{Z}^{(m)}_{n_o-1}, \mathcal{Z}^{(m)}_{n_o}$ from the MCMC output
3: transform the seam states to the Gaussian scale by the PIT (4)–(6)
4: **for** $h = 1, \ldots, H$ **do**
5: draw $\mathbf{w}^*, z^*, \sigma^{2*}$ at lead h from the AR(2) recursion (28)–(30) at $\boldsymbol{\phi}^{(m)}$, with the innovation variances (14)
6: invert the PIT to obtain $\mathcal{Z}^{(m)}_{n_o+h}$
7: draw $\mathbf{Y}^{(m)}_{n_o+h}$ from the conditional Gaussian field (26) at $\mathcal{Z}^{(m)}_{n_o+h}$ and $\boldsymbol{\theta}^{(m)}$
8: **end for**
9: **end for**
10: **return** the predictive sample $\{\mathbf{Y}^{(m)}_{n_o+1:n_o+H}\}_{m=1}^{M}$

Line 4 of Algorithm 1 advances the knot locations along with the scale and skewness latents, and line 6 places every site in the subregion of (27) that its forecast knots define. At $K = 1$ the partition is the whole domain at every lead, so the knot locations leave the field unchanged and only z and σ^2 carry the temporal signal into the forecast. The forecast evaluation of Section 3.3 is set at $K = 1$ for that reason, which also gives the AR(2), AR(1), and independent fits a common partition to be compared on.

3 Simulation study

This simulation study examines whether, and under which data-generating settings, the AR(2) and MRTS extensions improve held-out exceedance prediction. Section 3.1 gives the design shared by all experiments and Section 3.2 the validation protocols, including the uncertainty attached to the paired score comparisons. Section 3.3 evaluates the AR(2) extension under autoregressive and long-memory temporal dynamics; Section 3.4 evaluates the MRTS extension under second-moment and first-moment spatial non-stationarity. Throughout, both extensions are assessed as candidate options, benchmarked against Gaussian fits and against matched models without each extension.

3.1 Common simulation design

All simulation designs generate data from the hierarchical model of Section 2.5 on $n_s = 144$ sites, drawn uniformly on $[0, 10] \times [0, 10]$, over $n_t = 50$ days ($n_t = 200$ in the time-block forecast evaluation of Section 3.3). The two nonlinear-mean designs of Section 3.4 are the single exception, replacing the skew-t error process with a Gaussian one. In every design the Matérn correlation (2) is set to $\eta = 0.9$, $\nu = 1/2$, and $\varphi = 1$. In the data generation the regression mean is held constant at $\mathbf{X}^\top\boldsymbol{\beta} = 10$, while every fitted model includes an intercept and a first-order spatial trend. The designs of Section 3.4 depart from that constant mean in two ways. One adds a mean-zero non-stationary random effect, which leaves the regression mean at 10; the two Gaussian-error designs carry a fixed nonlinear mean surface in its place. For each design we simulate 10 data sets and run every MCMC chain for $20{,}000$ iterations, discarding the first $10{,}000$ as burn-in. The spatial hold-out evaluation splits each data set into 100 training sites and 44 withheld test sites, while the time-block forecast evaluation observes all 144 (Section 3.2). Throughout, $q(0.80)$ denotes the 80th sample quantile of the data, and a threshold $T = q(0.80)$ means censoring the response there. The five baseline analysis models are

1. Gaussian marginal, $K = 1$ knot;
2. Skew-t marginal, $K = 1$ knot, no thresholding;
3. Symmetric-t marginal, $K = 1$ knot, threshold $T = q(0.80)$;
4. Skew-t marginal, $K = 5$ knots, no thresholding;
5. Symmetric-t marginal, $K = 5$ knots, threshold $T = q(0.80)$.

Censoring is paired with the symmetric-t marginal because the skewness parameter is only weakly identified once the lower tail is censored, following Morris, Reich, Emeric Thibaud, and Cooley (2017).

3.2 Validation protocols and uncertainty

Spatial hold-out validation. Each setting is fit to the 100 training sites of a replicate and scored at its 44 withheld sites. Because the target is exceedance of a high level, we compare models by the Brier score (Gneiting and Raftery, 2007). For an exceedance level L, the Brier score is

$$\mathrm{BS}(L) \;=\; \left\{\, I(y > L) \;-\; \widehat{P}(Y > L) \,\right\}^2, \tag{34}$$

where y is the held-out observation, $I(y > L)$ is the exceedance indicator, and $\widehat{P}(Y > L)$ is the posterior predictive probability of exceeding L. We average the Brier scores over all test sites and days; lower values indicate better exceedance prediction, and L ranges over the sample quantiles reported with each figure.

Time-block forecast validation. The site-splitting scheme above leaves the day being predicted observed at the training sites, so its predictive distribution keeps the kriging term of Section 2.6 and borrows from that day. The time-block forecast evaluation of Section 3.3 withdraws that support and scores the forecast of Algorithm 1 instead. All 144 sites are observed up to a forecast origin, the next 15 days are predicted at those same sites, and no observation after the origin is available anywhere. Those 15 days are one forecast window. Each data set supplies five origins, at days 50, 80, 110, 140 and 170 of a 200-day record, so the evaluation does not rest on a single stretch of the record. For each forecast lead h the Brier score (34) is computed with L set to the empirical quantile of that data set's whole record, which is the rule the spatial hold-out already uses, so the two evaluations are measured on one scale. Results are reported at the 0.90 and 0.95 quantiles and over two lead windows, leads 1–5 and leads 1–15. Scores are averaged over sites, over the leads of the reported lead window, and over the five forecast windows, leaving one number per data set, and that number is the unit of the paired comparisons below.

Uncertainty of score comparisons. A mean Brier score is an estimate, and two such estimates computed on the same held-out sites are strongly positively correlated, since a site that is hard to predict is hard for every model. We therefore attach uncertainty to the *paired difference* of scores. For a matched pair of models A and B and an exceedance level L, the estimand is

$$\Delta(L) \;=\; \mathrm{E}\big[\mathrm{BS}_A(L)\big] \;-\; \mathrm{E}\big[\mathrm{BS}_B(L)\big], \tag{35}$$

estimated by forming the difference site by site on the shared held-out set and assessing its sampling variability over the 10 simulated data sets. Each comparison tests

$$H_0\colon\ \Delta(L) = 0 \qquad \text{against} \qquad H_1\colon\ \Delta(L) \neq 0, \tag{36}$$

summarized by a 95% paired-t confidence interval and a Wilcoxon signed-rank test, which does not assume normal differences; with 10 replicates the Wilcoxon p-values move on a coarse grid whose smallest attainable two-sided value is $2^{-9} \approx 0.002$. Each pair varies only the factor under test, with the analysis knot count matched to the generating design. In the tables that follow, a star marks an interval that excludes zero. Among the dozens of paired comparisons reported in this thesis, an isolated borderline star is within what chance alone would produce. An interval that covers zero means the data fail to reject H_0, not that the two models have been shown equal.

3.3 The AR(2) experiment

Design. To test the AR(2) analysis model against data with a known autoregressive structure, the six generating designs are skew-t ($\lambda = 3$) with temporal dependence imposed on the latent σ^2, $\mathbf{w}$, and z processes. The first five evolve through a fixed AR(2) recursion (16)–(18); the sixth replaces it with a long-memory ARFIMA$(0, d, 0)$ process:

1. $K = 1$ knot, strong dependence $(\phi_1, \phi_2) = (0.80, -0.35)$;
2. $K = 1$ knot, weak dependence $(\phi_1, \phi_2) = (0.12, -0.05)$;
3. $K = 5$ knots, strong dependence $(\phi_1, \phi_2) = (0.80, -0.35)$;
4. $K = 5$ knots, weak dependence $(\phi_1, \phi_2) = (0.12, -0.05)$;
5. $K = 1$ knot, AR(2) $(\phi_1, \phi_2) = (0.15, 0.80)$;
6. $K = 1$ knot, long-memory ARFIMA$(0, d, 0)$, $d = 0.45$ (Davies and Harte, 1987).

All three AR(2) coefficient pairs lie inside the stationarity triangle $\mathcal{S}$. Every design is fit by the same nine analysis models, namely the five baselines and the skew-t models with AR(1) and AR(2) coefficients at $K = 1$ and $K = 5$ knots.

Designs 5 and 6 are evaluated under both protocols of Section 3.2. The spatial hold-out evaluation fits the same nine models as the other designs; the time-block forecast evaluation fits the three $K = 1$ skew-t variants whose latent processes are i.i.d. in time, AR(1), and AR(2), so that the value of the second lag is read off directly as the AR(2) minus AR(1) contrast. The fixed-AR(2) design with $(\phi_1, \phi_2) = (0.80, -0.35)$ joins them as a short-memory control, fit by the same three variants. Its autocorrelation decays within six days, well inside the 15-day forecast horizon, so a comparison that reports a gain there would be reporting the extra parameters rather than the memory.

The forecast fits differ from the rest of the study in one respect, and for an identifiability reason. The mean of (1) involves λ and $|z_t(\mathbf{s})|$ only through their product, so the data identify the pair only up to a positive scale factor, and it is the half-normal law on z that pins the scale down. On the short prefixes that a forecast origin leaves, that law is weakly informative, and a chain can settle on a reflected branch in which λ turns negative and β_0 rises so that the fitted mean is unchanged, while the predictive standard deviation grows in proportion to $|\lambda|$. The forecast fits therefore restrict the support, $\lambda \sim \mathrm{HN}(0, 20)$, which fixes the direction of the skewness without constraining its magnitude and is drawn exactly by a truncated Gibbs step. Right skewness is the case of interest both in these designs, generated at $\lambda = 3$, and in the ozone data of Section 4.1, so fixing the sign discards no case that the evaluation needs. The restriction is confined to the time-block forecast fits of this experiment. Every other fit reported in this thesis keeps the unrestricted $N(0, 20)$ prior of (31), including the spatial hold-out fits of this experiment, those of Section 3.4, and the ozone fits of Section 4. The reflected branch is a property of the model rather than of the forecast protocol, so it can arise under the unrestricted prior as well. What differs is the information available to rule it out. A spatial hold-out fit sees all 50 days at 100 sites, so the scale of z is far better determined there than on the prefix a forecast origin leaves, and the restriction is needed only where that information is thin.

Results. In the spatial hold-out evaluation, predictive ranking is governed by the marginal and the spatial partition. The methods whose marginal matches the generating truth give the largest improvement over the Gaussian model, a mismatched marginal can perform worse than Gaussian in the far tail, and the preferred knot count tracks the generating truth (Figure 1(a)–(f)). The temporal structure barely moves these scores. The AR(2) and AR(1) curves are essentially indistinguishable from their non-temporal counterparts at every quantile and in every setting, and the paired intervals of Table 2 confirm the visual impression. Of the 36 intervals, 35 cover zero, and the single starred exception ($K = 5$, $(0.80, -0.35)$, at the 0.90 quantile) favors the *non-temporal* baseline and sits at the edge of significance, within what

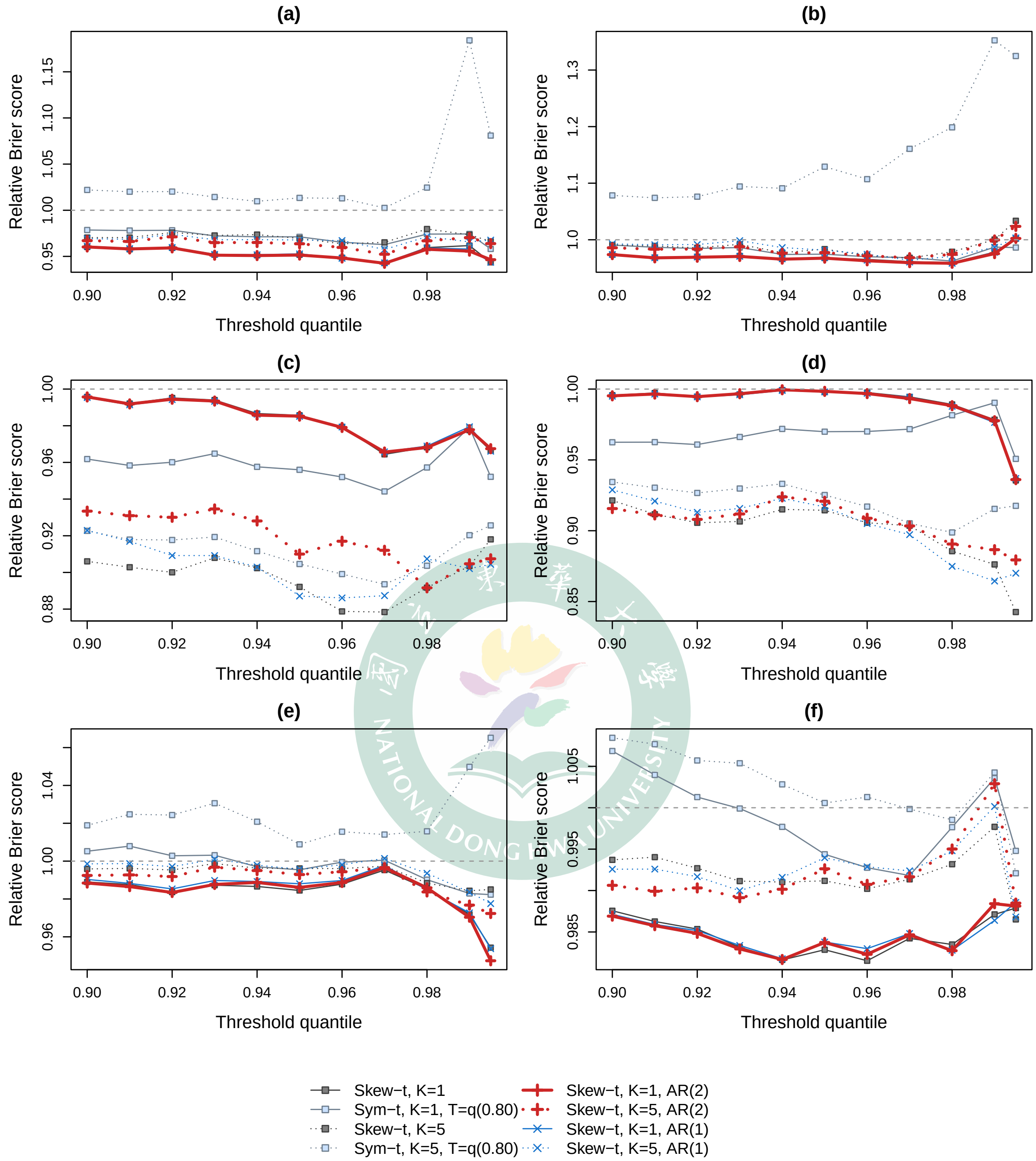


Figure 1: Spatial hold-out evaluation: relative Brier score versus the Gaussian model by threshold quantile (Gaussian is the reference $\equiv 1$, shown as the dashed line; values below it are better). All data-generating settings are skew-t with $\lambda = 3$: (a) $K = 1$, AR(2) $(\phi_1, \phi_2) = (0.80, -0.35)$; (b) $K = 1$, AR(2) $(0.12, -0.05)$; (c) $K = 5$, AR(2) $(0.80, -0.35)$; (d) $K = 5$, AR(2) $(0.12, -0.05)$; (e) $K = 1$, AR(2) $(0.15, 0.80)$; (f) $K = 1$, long memory, ARFIMA$(0, d, 0)$ with $d = 0.45$.

chance alone produces among this many comparisons. The two-sided Wilcoxon p-values, ranging from 0.05 to 1.00, agree.

Table 2: Spatial hold-out evaluation of the AR(2) experiment: paired Brier differences over the 10 data sets, $\Delta = \mathrm{BS}_A - \mathrm{BS}_B$ with 95% paired-t confidence intervals. A star marks an interval that excludes zero.

		$\Delta \times 10^3$ [95% CI]	
Generating design	L quantile	AR(2) − i.i.d.	AR(2) − AR(1)
K=1, $(0.80, -0.35)$	0.90	$+0.00$ $[-0.03, +0.04]$	-0.03 $[-0.06, +0.00]$
	0.95	-0.01 $[-0.03, +0.01]$	-0.02 $[-0.06, +0.02]$
	0.98	-0.01 $[-0.04, +0.01]$	-0.01 $[-0.03, +0.01]$
K=1, $(0.12, -0.05)$	0.90	$+0.00$ $[-0.02, +0.03]$	-0.01 $[-0.04, +0.02]$
	0.95	-0.00 $[-0.03, +0.02]$	-0.01 $[-0.03, +0.01]$
	0.98	-0.01 $[-0.03, +0.01]$	$+0.00$ $[-0.01, +0.01]$
K=5, $(0.80, -0.35)$	0.90	$+0.93^{*}$ $[+0.12, +1.75]$	$+0.36$ $[-0.51, +1.23]$
	0.95	$+0.32$ $[-0.33, +0.97]$	$+0.41$ $[-0.12, +0.94]$
	0.98	-0.00 $[-0.33, +0.32]$	-0.14 $[-0.33, +0.05]$
K=5, $(0.12, -0.05)$	0.90	-0.21 $[-1.16, +0.74]$	-0.48 $[-1.16, +0.20]$
	0.95	$+0.13$ $[-0.27, +0.52]$	$+0.08$ $[-0.34, +0.50]$
	0.98	$+0.04$ $[-0.26, +0.35]$	$+0.13$ $[-0.37, +0.63]$
K=1, $(0.15, 0.80)$	0.90	-0.02 $[-0.07, +0.03]$	-0.07 $[-0.23, +0.09]$
	0.95	$+0.04$ $[-0.00, +0.08]$	-0.05 $[-0.19, +0.10]$
	0.98	$+0.01$ $[-0.02, +0.03]$	$+0.01$ $[-0.02, +0.04]$
ARFIMA, $d = 0.45$	0.90	-0.03 $[-0.08, +0.01]$	-0.01 $[-0.04, +0.02]$
	0.95	$+0.02$ $[-0.01, +0.06]$	-0.00 $[-0.03, +0.03]$
	0.98	-0.01 $[-0.06, +0.04]$	-0.00 $[-0.04, +0.03]$

Coefficient recovery separates the two possible readings of this insensitivity (Table 3). In the first four designs the temporal signal is only weakly identified. The scale process is the one component recovered with fidelity, and only where the generating design carries a single knot under strong dependence, at 0.803 and -0.363 against a truth of 0.80 and -0.35. Five generating knots split that signal among five localized scale processes and the estimates attenuate toward zero, to 0.512 and 0.043 under a one-knot fit and to 0.169 and -0.023 under a matched five-knot fit. Under weak dependence the posterior means sit near the true values, at 0.083 and -0.019 against 0.12 and -0.05, but their spread across data sets, 0.122 and 0.150, exceeds the estimates themselves, so nothing there is separated from zero. The skewness process repeats the same pattern attenuated, and the spatial process recovers nothing wherever it is fit.

Designs 5 and 6 are where the temporal signal is identified. Under $(\phi_1, \phi_2) = (0.15, 0.80)$ the matched fit recovers the generating dynamics closely, in particular the dominant second lag that an AR(1) can only compress into a single lag, at 0.742 on the scale process and 0.728 on the skewness process against a truth of 0.80. Under long memory the fit registers persistence but loads it on the first lag, at 0.502 against a second lag of 0.200 whose spread of 0.239 across data sets leaves it undetermined. Yet the hold-out scores of these two designs repeat the null pattern in sharper form (Figure 1(e)–(f); Table 2, lower block). Identified temporal parameters with unmoved scores point to the task rather than the model, exactly what the protocol distinction of Section 3.2 anticipates. When the day being predicted is already observed at the training sites, that day's cross-sectional information dominates and the temporal prior has little left to add.

Table 3: Recovered AR(2) coefficients of the scale, skewness, and knot-location processes. Each entry is the posterior mean averaged over the 10 data sets, and the figure in parentheses is the standard deviation of those 10 posterior means, a measure of how far the estimate moves from one data set to the next rather than of the posterior uncertainty within a single fit. A dash marks a coefficient the fit does not estimate.

Settings		Fit	$\hat{\phi}^{(\sigma)}$		$\hat{\phi}^{(z)}$		$\hat{\phi}^{(w)}$	
(ϕ_1, ϕ_2)	K	K	$\hat{\phi}_1$	$\hat{\phi}_2$	$\hat{\phi}_1$	$\hat{\phi}_2$	$\hat{\phi}_1$	$\hat{\phi}_2$
$(0.80, -0.35)$	1	1	0.803(0.140)	-0.363(0.137)	0.626(0.114)	-0.225(0.157)	-	-
$(0.80, -0.35)$	1	5	0.871(0.142)	-0.438(0.164)	0.472(0.284)	-0.270(0.162)	-0.129(0.377)	0.062(0.257)
$(0.12, -0.05)$	1	1	0.083(0.122)	-0.019(0.150)	0.186(0.137)	0.084(0.226)	-	-
$(0.12, -0.05)$	1	5	0.095(0.141)	-0.050(0.220)	0.050(0.088)	0.051(0.142)	0.002(0.215)	0.179(0.259)
$(0.80, -0.35)$	5	1	0.512(0.190)	0.043(0.245)	0.434(0.185)	-0.108(0.128)	-	-
$(0.80, -0.35)$	5	5	0.169(0.115)	-0.023(0.118)	0.074(0.093)	-0.006(0.090)	-0.019(0.206)	0.094(0.135)
$(0.12, -0.05)$	5	1	0.285(0.228)	0.161(0.164)	0.163(0.122)	-0.063(0.122)	-	-
$(0.12, -0.05)$	5	5	-0.001(0.072)	-0.017(0.137)	0.010(0.075)	-0.010(0.067)	-0.017(0.193)	0.115(0.190)
$(0.15, 0.80)$	1	1	0.112(0.132)	0.742(0.117)	0.175(0.152)	0.728(0.112)	-	-
ARFIMA, d=0.45	1	1	0.502(0.237)	0.200(0.239)	0.365(0.249)	0.156(0.169)	-	-

The time-block forecast evaluation answers the question the site split cannot, and it answers it differently for each generating design, according to whether the generating memory reaches past the forecast horizon in a form the AR(2) prior can represent (Figure 2, Table 4). Under $(\phi_1, \phi_2) = (0.15, 0.80)$ the AR(2) fit improves on both competitors at both thresholds, at both lead windows, and at every individual lead. Pooled over the full horizon its mean Brier score is 0.0949 against 0.1207 for the i.i.d. fit at the 0.90 quantile, a reduction of 21% with a paired interval of $[-42.5,\ -9.1] \times 10^{-3}$; at the 0.95 quantile the reduction is 13% with

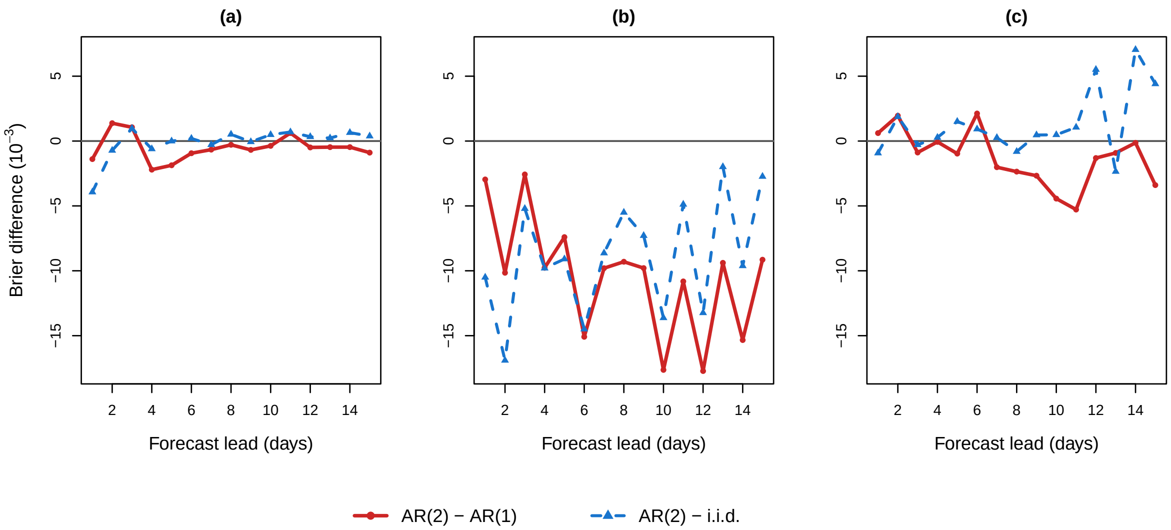


Figure 2: Time-block forecast evaluation of the AR(2) experiment: paired difference in Brier score at the 0.95 threshold quantile by forecast lead, in units of 10^{-3}, negative favoring the AR(2) fit. Generating designs: (a) AR(2) $(\phi_1, \phi_2) = (0.80, -0.35)$; (b) AR(2) $(0.15, 0.80)$; (c) long memory, ARFIMA$(0, d, 0)$ with $d = 0.45$.

the interval $[-14.2,\ -3.6] \times 10^{-3}$. What pays there is the second lag rather than temporal pooling as such. The AR(1) fit is indistinguishable from the i.i.d. one over the full horizon, at $+1.6 \times 10^{-3}$ with an interval covering zero, whereas AR(2) minus AR(1) is -10.5×10^{-3} with the interval $[-16.0,\ -5.0] \times 10^{-3}$, and that contrast separates at every threshold from 0.90 to 0.98. The short-memory control fixes the scale of the claim. Its autocorrelation decays within six days, so over a fifteen-day horizon nothing should be detectable, and nothing is. The pooled AR(2) minus i.i.d. difference is -0.00×10^{-3} at the 0.90 quantile and -0.08×10^{-3} at the 0.95, both intervals covering zero, and the same holds at every one of the eleven thresholds. One design in which the extra lag is worth nothing returns nothing, and one in which it is worth a great deal returns a large and well separated gain, under a single protocol and a single estimator.

Under long memory no contrast separates from zero. No interval, at either threshold or either lead window, excludes zero, and no difference in the block exceeds 2.1×10^{-3} in magnitude. A calibration check of the predictive tail locates the reason. Alongside each Brier score we record the predicted exceedance probability and the realized exceedance frequency at the same level, both computed from the forecasts and the thresholds that enter Table 4 and both averaged over the held-out sites, the fifteen leads, the five forecast windows and the ten data sets. Under long memory the AR(2) predictive over-predicts exceedance at every level of the eleven-threshold grid, and the ratio of the predicted rate to the realized one rises monotonically with the level. It runs from 1.13 at the 0.90 quantile, where 0.1250 is predicted against a realized 0.1111, to 4.11 at the 0.995 quantile, the deepest level of the grid, where 0.0234 is predicted against a realized 0.0057. The fitted AR(2) therefore misplaces the upper-tail level of its predictive distribution under hyperbolic decay, and a temporal structure that is fitted but mislevelled cannot convert into skill at a threshold. The mean scores are ordered consistently with that reading. Over the full horizon the AR(1) fit carries the highest mean Brier score at both thresholds, so a single geometric lag is the least appropriate summary of a hyperbolically decaying memory. The three fits nevertheless lie within 2.7×10^{-3} of one another, inside the width of every interval in the block, so that ordering is descriptive and no claim rests on it.

Of the 24 intervals in Table 4, 8 exclude zero and all 8 belong to the $(0.15, 0.80)$ design. The smallest two-sided p-value is 0.0004, which survives a Bonferroni correction across all 24 comparisons at $0.05/24 = 0.002$, so that result is not an artifact of multiplicity. The two-sided Wilcoxon p-values, ranging from 0.002 to 1.00, agree with the intervals throughout. Read beside the spatial hold-out evaluation, which is null under the same generating designs and the same models, the contrast is a property of the prediction task. Temporal structure is worth little when the day in question is observed elsewhere, and worth a great deal when it is not.

3.4 The MRTS experiment

Design. This experiment enriches the regression component with the MRTS basis functions $f_1(\mathbf{s}), \ldots, f_{K_{\text{MRTS}}}(\mathbf{s})$ of Section 2.4. The four generating designs place the non-stationarity first where the mean basis cannot reach it (designs 1–2) and then where it can (designs 3–4):

1. Skew-t , $K = 5$ knots, $\lambda = 3$, with the constant mean $\mathbf{X}^\top \boldsymbol{\beta} = 10$;

2. Skew-t , $K = 1$ knot, $\lambda = 3$, with an additive mean-zero non-stationary spatial random effect $u_t(\mathbf{s})$, redrawn daily from two fixed cosine fields (Appendix D), so the non-stationarity lies in the second moment while the marginal mean stays at 10;

3. Gaussian, sigmoid front: the nonlinear-mean model (37) with $\tilde{g}(\mathbf{s}) = \tanh\{\|\mathbf{s} - (2,2)^\top\| - 4.5\}$, a smoothed circular step;

Table 4: Time-block forecast evaluation of the AR(2) experiment: mean Brier score by lead window at the two threshold quantiles, and paired differences with 95% paired-t confidence intervals over the 10 data sets (Δ in units of 10^{-3}; negative favors the first-named model; a star marks an interval that excludes zero). The fitted models are the skew-t $K = 1$ variants with i.i.d., AR(1), and AR(2) latent temporal structure.

		Mean Brier score			$\Delta \times 10^3$ [95% CI]	
Generating design	Leads	i.i.d.	AR(1)	AR(2)	AR(2)−i.i.d.	AR(2)−AR(1)
Threshold quantile 0.90						
AR(2) $(0.80, -0.35)$	1–5	0.0852	0.0847	0.0843	-0.94 $[-8.25, +6.36]$	-0.45 $[-4.82, +3.92]$
	1–15	0.0875	0.0883	0.0875	-0.00 $[-2.52, +2.52]$	-0.79 $[-3.12, +1.54]$
AR(2) $(0.15, 0.80)$	1–5	0.1060	0.0931	0.0740	-31.99^* $[-53.22, -10.75]$	-19.12^* $[-30.89, -7.35]$
	1–15	0.1207	0.1188	0.0949	-25.80^* $[-42.50, -9.10]$	-23.89^* $[-33.65, -14.12]$
ARFIMA, $d = 0.45$	1–5	0.0982	0.0967	0.0976	-0.63 $[-6.13, +4.88]$	$+0.94$ $[-3.84, +5.72]$
	1–15	0.1024	0.1043	0.1022	-0.14 $[-5.02, +4.74]$	-2.08 $[-8.80, +4.63]$
Threshold quantile 0.95						
AR(2) $(0.80, -0.35)$	1–5	0.0505	0.0502	0.0496	-0.86 $[-5.15, +3.43]$	-0.61 $[-3.07, +1.86]$
	1–15	0.0467	0.0471	0.0466	-0.08 $[-1.48, +1.33]$	-0.51 $[-1.72, +0.70]$
AR(2) $(0.15, 0.80)$	1–5	0.0556	0.0518	0.0453	-10.30^* $[-15.94, -4.66]$	-6.57^* $[-11.57, -1.58]$
	1–15	0.0680	0.0696	0.0591	-8.90^* $[-14.19, -3.61]$	-10.46^* $[-15.97, -4.95]$
ARFIMA, $d = 0.45$	1–5	0.0529	0.0532	0.0534	$+0.49$ $[-2.28, +3.26]$	$+0.13$ $[-2.19, +2.45]$
	1–15	0.0556	0.0583	0.0569	$+1.30$ $[-1.54, +4.13]$	-1.32 $[-5.98, +3.34]$

4. Gaussian, transform mix: the nonlinear-mean model (37) with $\tilde{g}(\mathbf{s}) = \text{std}\{\log(1+s_1)\} + \text{std}\{e^{-s_1/2}\} + \text{std}\{s_1/(1+s_2)\}$, where $\text{std}\{\cdot\}$ standardizes each term to unit spatial standard deviation. The curvature of this surface concentrates near the domain boundary, where the uniform site draw is sparsest.

The nonlinear-mean designs replace the skew-t generator with Gaussian errors,

$$Y_t(\mathbf{s}) \;=\; 10 \;+\; g(\mathbf{s}) \;+\; \sigma_g\,\varepsilon_t(\mathbf{s}), \qquad \sigma_g = 4, \tag{37}$$

where the errors ε_t are standard Gaussian processes with the Matérn correlation (2) at the generating values of Section 3.1, drawn independently across days; every t-family analysis model is therefore misspecified in its error law. Each raw surface $\tilde{g}$ is orthogonalized against the baseline design $[1, s_1, s_2]$ and standardized to a spatial standard deviation of 11, so the $K_{\text{MRTS}} = 0$ design spans none of the resulting g.

Each of the five baseline models is fit across a grid of MRTS basis sizes, $K_{\text{MRTS}} \in \{0, 3, 5, 8, 10, 12, 15, 20, 25\}$, where $K_{\text{MRTS}} = 0$ recovers the no-MRTS baseline. On both nonlinear-mean designs the grid is extended by $\{30, 40, 50\}$, since the finer surfaces are resolved only by a fine basis.

Results. On the two skew-t -core designs, augmenting the regression component with MRTS basis functions does not deliver a stable improvement in upper-tail scores, and enlarging the basis does not lower the average Brier score on either setting (Figure 3). On the five-knot setting every curve is essentially flat in K_{MRTS}; on the additive non-stationary random-effect process the curves are likewise flat, with over-specified models destabilizing at large K_{MRTS}. As in the AR(2) experiment, the generating knot count dominates the ranking. The second setting is the decisive one. The non-stationarity lives in the second moment (a daily-redrawn random effect u_t), whereas the MRTS basis enters only the fixed-effect mean. Because u_t has zero time-average, the time-pooled coefficient $\boldsymbol{\beta}^*$ has no first-moment structure to recover, so the Brier curves stay flat by construction rather than through absence of signal, since u_t contributes substantial (spatially non-stationary) variance. The MRTS mean term therefore yields no reliable tail-score gain. A fixed-effect mean cannot represent a non-stationary second-moment feature, and so, in these designs, the spatial mean is not the binding constraint on upper-tail prediction.

The two nonlinear-mean designs address the complementary case (Figure 4; Table 5). On the sigmoid-front design the Brier score finally responds to the mean basis, but only once the

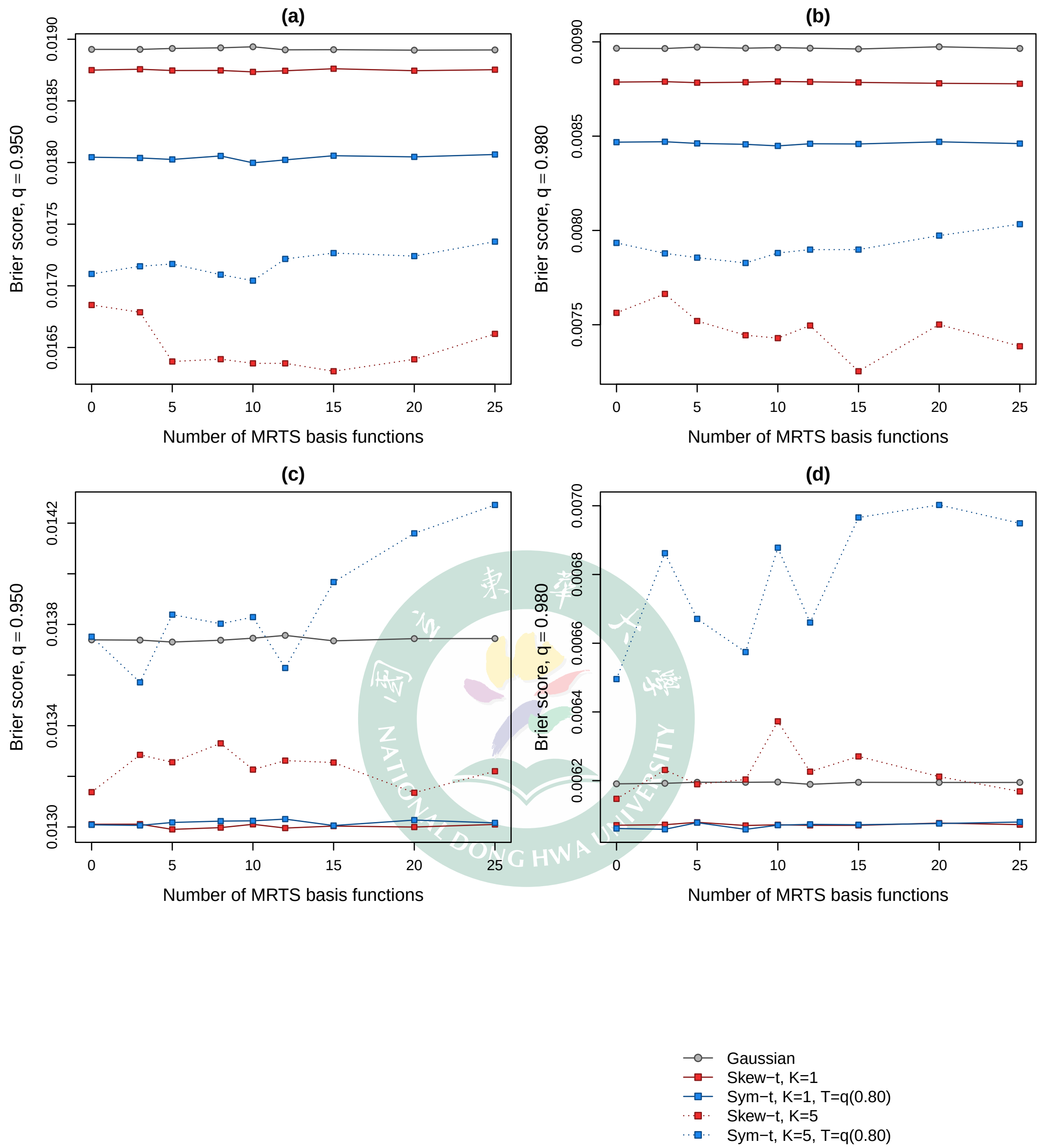


Figure 3: Brier score versus the number of MRTS basis functions K_{MRTS} ($K_{\text{MRTS}} = 0$ is the no-MRTS baseline; lower is better). Data-generating processes: (a,b) skew-t ($K = 5$, $\lambda = 3$); (c,d) skew-t ($K = 1$, $\lambda = 3$) with an additive non-stationary spatial random effect. Exceedance level $q = 0.95$ in (a,c) and $q = 0.98$ in (b,d).

basis is fine enough to resolve the front. The profiles stay flat for a coarse basis, deteriorate slightly at intermediate sizes, and improve only at the finer bases, with the gains larger at the higher exceedance level. The deterioration at intermediate sizes has a single cause. A partially fitted mean sharpens the predictive distribution while leaving a residual surface error that carries over to the held-out sites, and the Brier score worsens until the basis actually resolves the front.

The transform-mix design shows the same mechanism operating from a much worse starting point. The mismatched five-knot partition, compounded for the thresholded variant by censoring, leaves the no-basis baselines of the five-knot models far above those of the one-knot models, and the mean basis recovers most of that loss. Every model improves at the 0.95 threshold quantile, and at its best basis size each converges to nearly the same score. The improvement is not uniform across levels, however. At the rarer 0.98 quantile it is carried entirely by the five-knot models, whereas the one-knot models stay flat or worsen. This is the sharpness/residual-error signature again, in that only the models whose no-basis fit is badly misspecified, here the five-knot partition, carry enough headroom for the mean basis to convert into a gain at the rarest level. Both surfaces leave genuine Brier headroom. Replacing the fitted mean by the true surface g under the true Gaussian error law improves the score substantially on both. The realized gains convert only a small part of that headroom, which is consistent with the sharpness/residual-error account above. The Brier score improves only where the residual basis error is small relative to the predictive spread.

Table 5: Nonlinear-mean designs: Brier score at the 0.95 threshold quantile relative to the same model's own $K_{\text{MRTS}} = 0$ fit (lower is better). Best is the smallest ratio over $K_{\text{MRTS}} > 0$, with the minimizing basis size in parentheses; BS_0 is the absolute Brier score at $K_{\text{MRTS}} = 0$; thr. denotes the thresholding $T = q(0.80)$.

Generating design	Analysis model	K_{MRTS}=30	K_{MRTS}=40	K_{MRTS}=50	Best (K_{MRTS})	BS_0
Sigmoid front	Gaussian	0.970	0.967	0.966	0.966 (50)	0.0232
Sigmoid front	Skew-t , $K = 1$	0.969	0.967	0.967	0.966 (25)	0.0233
Sigmoid front	Symmetric-t, $K = 1$, thr.	0.943	0.949	0.949	0.941 (25)	0.0240
Sigmoid front	Skew-t , $K = 5$	0.960	0.956	0.956	0.956 (50)	0.0235
Sigmoid front	Symmetric-t, $K = 5$, thr.	0.944	0.951	0.974	0.944 (25)	0.0239
Transform mix	Gaussian	0.708	0.695	0.679	0.679 (50)	0.0079
Transform mix	Skew-t , $K = 1$	0.698	0.686	0.669	0.669 (50)	0.0080
Transform mix	Symmetric-t, $K = 1$, thr.	0.554	0.576	0.679	0.554 (30)	0.0095
Transform mix	Skew-t , $K = 5$	0.303	0.264	0.257	0.257 (50)	0.0210
Transform mix	Symmetric-t, $K = 5$, thr.	0.374	0.446	0.367	0.367 (50)	0.0131

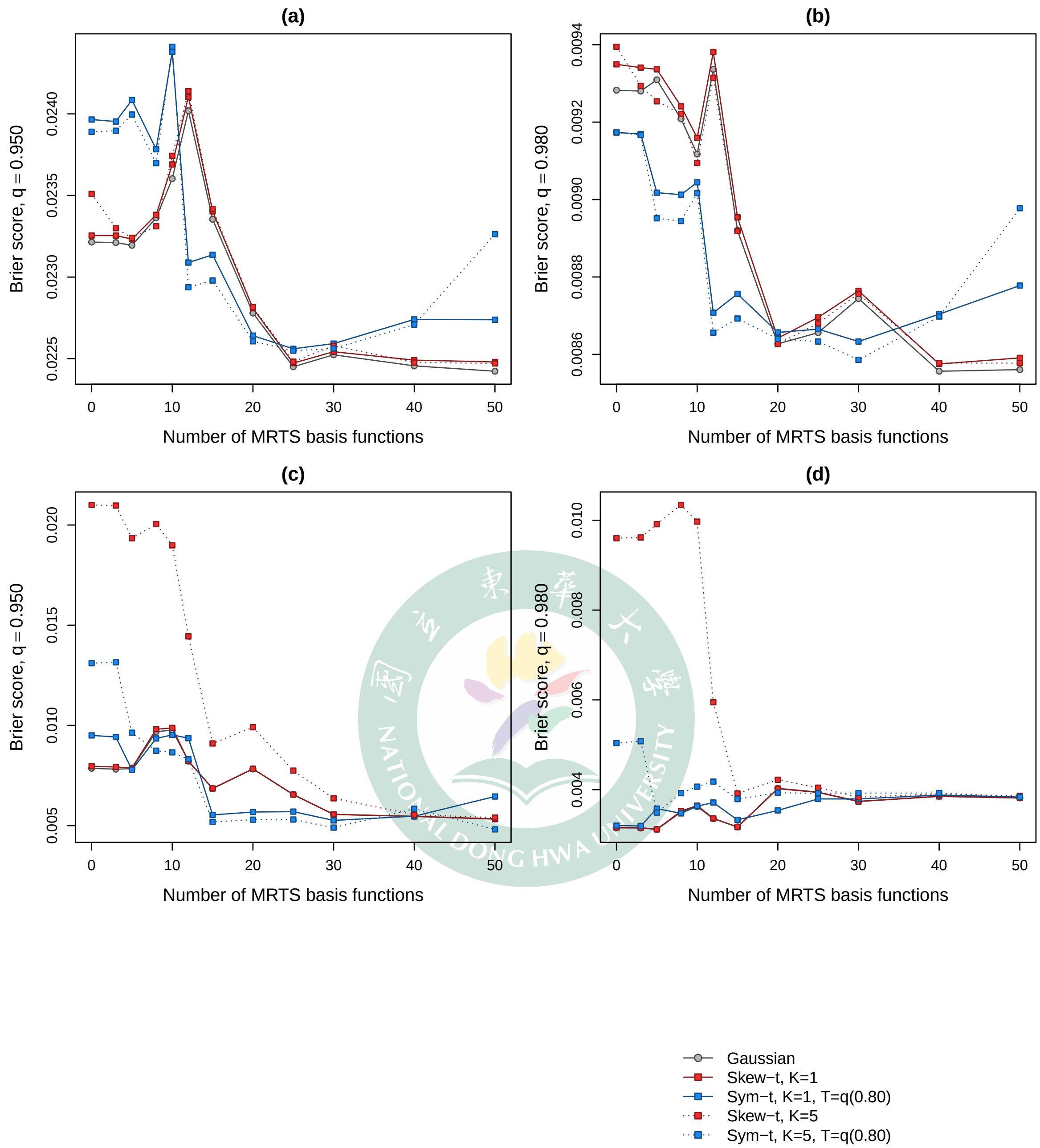


Figure 4: Brier score versus the number of MRTS basis functions K_{MRTS} on the two nonlinear-mean designs of (37) ($K_{\text{MRTS}} = 0$ is the no-MRTS baseline; lower is better). Data-generating processes: (a,b) sigmoid front; (c,d) transform mix. Exceedance level $q = 0.95$ in (a,c) and $q = 0.98$ in (b,d).

3.5 Summary of the simulation study

Overall, the simulation study indicates that both extensions enlarge the usable model class but that neither uniformly dominates the STP baseline. The AR(2) experiment sharpens this conclusion into a statement about the prediction task itself. Even under the most persistent generating design, $(\phi_1, \phi_2) = (0.15, 0.80)$, where the temporal coefficients are recovered closely, prediction at withheld sites is insensitive to the autoregressive order, and the same holds under long memory. The gains from richer temporal pooling appear only once the task is to forecast days beyond the end of the record, and there they turn on whether the generating memory reaches past the forecast horizon in a form the AR(2) prior can represent. Under $(\phi_1, \phi_2) = (0.15, 0.80)$ it does, and the i.i.d. score falls by 13 to 21% depending on the threshold. Under $(0.80, -0.35)$ the autocorrelation decays inside the horizon and there is nothing left to gain, while under hyperbolic decay the fitted AR(2) misplaces the upper tail of its predictive distribution and no gain appears either. Whatever promise the AR(2) extension retains therefore lies in forecasting rather than in prediction at withheld sites. The MRTS experiments admit an equally structural summary. The mean term is inert when the non-stationarity lives in the second moment, and under nonlinear mean structure it pays off only once the basis is fine enough to resolve the surface, converting a small fraction of the oracle headroom. Whether the same pattern holds on real environmental data is the question taken up in Section 4, where the AR(2) and MRTS variants enter as candidate options to be selected by held-out predictive scores rather than as default replacements for the AR(1) STP baseline.

4 Ozone data analysis

Morris, Reich, Emeric Thibaud, and Cooley (2017) introduced the spatio-temporal skew-t (STP) process to predict high-threshold ground-level ozone exceedances from U.S. EPA monitoring data. We return to the same July 2005 data, asking not whether a new model class is needed but whether enlarging the STP candidate class along the two axes of this thesis, the AR(2) temporal prior and the MRTS spatial-mean term, yields better held-out tail prediction. The comparison is therefore designed to locate where in the upper tail, if anywhere, the enlarged candidate class holds its place among the leading models, rather than to select a single overall-best model.

4.1 Data

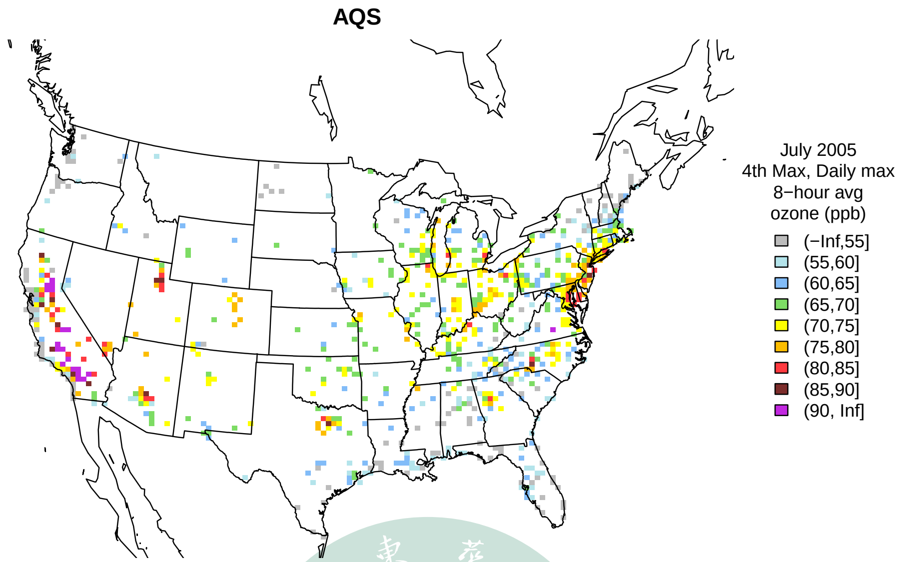
The response is the daily maximum 8-hour ozone concentration for the 31 days of July 2005 at U.S. EPA Air Quality System (AQS) monitoring sites, with the co-located Community Multi-scale Air Quality (CMAQ) model output as the sole covariate. Figure 5 summarizes the two data sources through the fourth-highest daily maximum 8-hour average of the month, the within-month analogue of the EPA design value. The AQS monitors are irregularly spaced, covering the eastern United States and California densely but leaving large gaps in the interior west, whereas CMAQ provides complete gridded coverage of the conterminous United States. CMAQ reproduces the broad spatial pattern of the observations but not their level at individual monitors, which motivates using it as a regression covariate to be calibrated rather than as a substitute for the observations. Following Morris, Reich, Emeric Thibaud, and Cooley (2017), the regression component is $\mathbf{X}_t(\mathbf{s}) = [1, \text{CMAQ}_t(\mathbf{s})]^\top$, so that each model acts as a statistical calibration of CMAQ against the monitor observations.

These data motivate the skew-t marginal directly. We regress the observations on CMAQ at the $1{,}089$ sites above and pool the residuals across sites and days. The QQ plot against the Normal distribution departs from the diagonal in both tails, with the upper tail by far the heavier (Figure 6), and the largest standardized residuals reach eight standard deviations, far beyond what a Gaussian model can accommodate. Against the skew-t distribution with skewness $\lambda = 1$ and $a = 10$ degrees of freedom that Morris, Reich, Emeric Thibaud, and Cooley (2017) use for the same diagnostic, the plot straightens except for the few most extreme points. The residual ozone variation is thus heavy-tailed and right-skewed exactly in the region that exceedance prediction cares about, and a skew-t process is a natural candidate class for it.

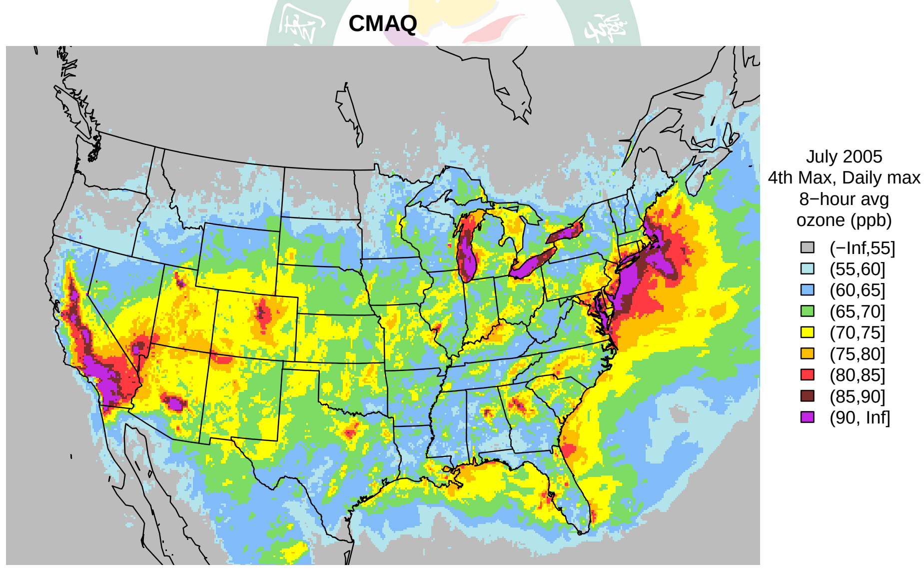
4.2 Candidate models and validation design

The baseline candidate set follows that of Morris, Reich, Emeric Thibaud, and Cooley (2017). We fit Gaussian and skew-t marginals over a range of partition sizes, $K \in \{1, 5, 6, 7, 8, 9, 10, 15\}$, and censor the response at $T = 0$, at $T = 50$ ppb (the 0.42 sample quantile), and at $T = 75$ ppb (the 0.92 sample quantile). For settings censored at $T = 75$ ppb, we use a symmetric-t marginal, as in the simulation study. Each setting is fit both with and without a latent time series.

On top of this baseline we add the two extensions developed in this thesis. The first replaces the AR(1) latent dynamics with the standardized AR(2) prior of Section 2.3; the second augments the regression component with MRTS basis functions, so that non-stationary spatial



(a) AQS monitor observations



(b) CMAQ model output

Figure 5: July 2005 fourth-highest daily maximum 8-hour ozone (the within-month analogue of the EPA design value, in ppb): (a) at the $1{,}089$ AQS monitoring sites missing at most 50% of the 31 days; (b) on the full CMAQ grid.

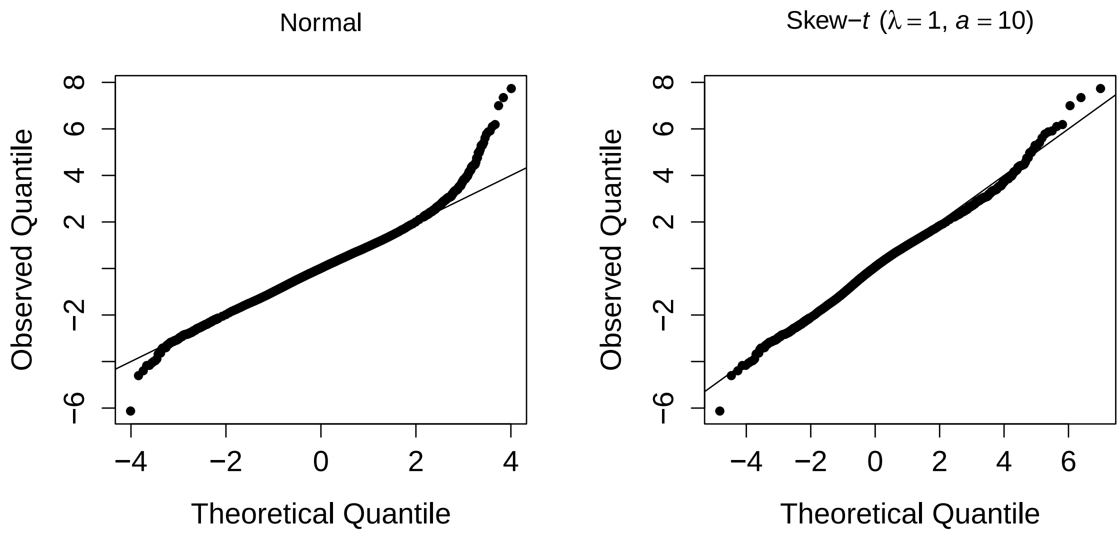


Figure 6: QQ plots of the pooled ozone-on-CMAQ regression residuals against the Normal (left) and the skew-t with $\lambda = 1$ and $a = 10$ (right) reference distributions, both standardized to zero mean and unit variance.

mean structure can be absorbed by multi-resolution thin-plate splines. Unlike the simulation study, which sweeps a grid of basis sizes, the ozone candidate set fixes $K_{\mathrm{MRTS}} = 10$, so the MRTS comparison below is a pre-specified one rather than a best-over-K_{MRTS} selection.

Of the $1{,}089$ sites of Figure 5, the comparison uses a stratified sample of 800, drawn to represent each region of the United States equally so that the interior west is not swamped by the dense eastern network. Models are evaluated by two-fold cross validation, with the 800 sites split at random into 400 training and 400 validation sites per fold, scored by the held-out Brier score of Section 3.2; robustness of the ranking to the split is examined with two further splits in Section 4.3. Both further splits draw on the same 800 sites, re-partitioned at random, and each is a single training and validation split rather than a pair of folds. One holds 300 training and 100 validation sites and the other 200 of each, so the remaining 400 sites of each partition are set aside and enter no score. Each MCMC chain is run for $30{,}000$ iterations with a burn-in period of $25{,}000$ iterations.

4.3 Results

Across the upper tail, the proposed extensions appear among the best models at scattered exceedance levels rather than improving prediction uniformly, so no single extension, and no single setting, is best across the whole tail (Table 6). The pattern is at least orderly. The MRTS extension leads at the 0.900 level, where the spatial mean still carries predictive signal; the AR(2) extension occupies both top ranks at 0.950 and returns at the deepest level, 0.995, where the best settings also shift toward high censoring and large partition sizes; the intermediate levels, 0.980 and 0.990, belong to the baseline family. At the 0.900 level the two leading settings are the MRTS fit and the otherwise identical fit without the basis, so the lead there is over the extension's own matched baseline rather than over an unrelated model. The extensions thus appear as the leading model at several levels; whether that standing survives a change of split is examined next.

Table 6: Top two models by held-out Brier score at exceedance level L, with the two-fold pooled score of each (lower is better).

Level L	Rank 1 model	BS	Rank 2 model	BS
0.900	**MRTS** skew-t , $K = 1, T = 0$, AR(1), $K_{\text{MRTS}} = 10$	0.044665	skew-t , $K = 1, T = 0$, AR(1)	0.044695
0.950	**AR2** skew-t , $K = 7, T = 50$	0.026640	**AR2** skew-t , $K = 10, T = 50$	0.026723
0.980	skew-t , $K = 6, T = 50$, AR(1)	0.012569	skew-t , $K = 5, T = 50$, AR(1)	0.012572
0.990	symmetric-t, $K = 9, T = 75$, AR(1)	0.006876	skew-t , $K = 5, T = 50$	0.006900
0.995	**AR2** symmetric-t, $K = 15, T = 75$	0.003842	symmetric-t, $K = 9$, $T = 75$, AR(1)	0.003851

Table 7: Top four Brier score models for the 300/100 split. Baseline denotes any other model carrying neither proposed extension; AR2 and MRTS denote the proposed temporal and spatial-mean extensions, and Both a setting that carries the two together. Rank 1 carries its absolute Brier score and ranks 2 to 4 the excess over it in units of 10^{-5}.

Level L	Rank 1	Rank 2	Rank 3	Rank 4
0.900	**MRTS** (0.04116)	Baseline (+16.5)	**MRTS** (+19.6)	Baseline (+23.2)
0.950	Baseline (0.01969)	Baseline (+0.8)	Baseline (+8.2)	Baseline (+12.5)
0.980	**MRTS** (0.00767)	**AR2** (+8.9)	Baseline (+13.3)	**AR2** (+14.5)
0.990	**AR2** (0.00309)	Baseline (+5.1)	Baseline (+6.6)	Baseline (+8.5)
0.995	Baseline (0.00127)	Baseline (+2.5)	Baseline (+2.9)	**AR2** (+3.4)

This picture is reproducible on smaller training samples. Repeating the comparison under two further train/validation splits, written $N_{\text{train}}/N_{\text{test}}$ for N_{train} training and N_{test} validation sites (the 300/100 and 200/200 splits; Tables 7 and 8), recovers the same qualitative ranking. The Baseline, AR(2), and MRTS families continue to share the top ranks across the tail, with two settings that carry both extensions among them, even as the exact best-ranked setting

Table 8: Top four Brier score models for the 200/200 split. Baseline denotes any other model carrying neither proposed extension; AR2 and MRTS denote the proposed temporal and spatial-mean extensions, and Both a setting that carries the two together. Rank 1 carries its absolute Brier score and ranks 2 to 4 the excess over it in units of 10^{-5}.

Level L	Rank 1	Rank 2	Rank 3	Rank 4
0.900	Baseline (0.05006)	**AR2** (+16.0)	Baseline (+28.7)	**Both** (+32.1)
0.950	**AR2** (0.03031)	**AR2** (+1.6)	**AR2** (+2.3)	Baseline (+3.9)
0.980	**Both** (0.01504)	**MRTS** (+1.3)	**AR2** (+1.6)	Baseline (+3.0)
0.990	Baseline (0.00771)	**AR2** (+1.2)	**Both** (+2.6)	Baseline (+2.9)
0.995	Baseline (0.00422)	Baseline (+0.2)	Baseline (+0.3)	**AR2** (+2.6)

shifts with the split and the level. That shifting is expected, since only a handful of validation sites exceed the deepest level. Behind the 0.995 score stand 120 exceedance events in the main split, 27 in the $200/200$ split and 2 in the $300/100$ split, so the fine ranking there is sensitive to the held-out draw.

What the ranking establishes should be read for what it is. The two leading scores at each level differ by between 3×10^{-6} and 8.3×10^{-5}. A single month of a single pollutant supplies one data set, so the replicate layer of Section 3.2 has no counterpart here. Uncertainty can still be attached at the site level. We resample the validation sites in clusters ($B = 2000$) and form the paired difference for each extension against its matched baseline, the AR(2) fit against its AR(1) counterpart and the MRTS fit against the same setting without the basis. Every AR(2) interval covers zero, on all three splits and at all five levels, and the sign of the difference changes from level to level. The MRTS differences are negative at every level of the two splits on which that pair was run, and three of them exclude zero, at the 0.950 and 0.990 levels of the main split and at the 0.995 level of the $200/200$ split, with two-sided bootstrap p-values of 0.011, 0.020 and 0.018. None of the three survives a correction for the ten comparisons that pair contributes, and no difference anywhere exceeds 1.5×10^{-4}. No extension is therefore established as the better predictor at any single level, though the consistent sign of the MRTS differences is what a small real gain would look like. What does carry across the tail, and across all three splits, is that both extensions hold their place among the leading models rather than falling behind it. The case for retaining them accordingly rests on the standing of the candidate class across splits and levels rather than on the size of any single gain.

5 Discussion

This thesis asked not whether the spatio-temporal skew-t process predicts high-threshold exceedances well, which Morris, Reich, Emeric Thibaud, and Cooley (2017) already established, but where within it room for sharper upper-tail prediction remains.

The temporal axis divides along the prediction task. Added memory leaves prediction at withheld sites unchanged, and not for want of signal. Under $(\phi_1, \phi_2) = (0.15, 0.80)$ the coefficients were recovered closely, and under long memory the fitted persistence was substantial. In both the held-out scores still did not move, because with the day in question observed at the training sites the cross-sectional information dominates. Forecasting days beyond the end of the record is where the memory paid, and what decided the payment was whether the generating memory reached past the horizon in a form the AR(2) prior could represent. It was nil where the autocorrelation decayed inside the forecast horizon, nil again under hyperbolic decay, where the fitted AR(2) misplaced the upper tail of its predictive distribution, and large under $(\phi_1, \phi_2) = (0.15, 0.80)$, where the mean Brier score fell by 13 to 21% against the i.i.d. fit and the whole of that gain was carried by the second lag rather than the first.

The spatial axis pays where the non-stationarity sits in the mean. A fixed-effect basis reaches first-moment structure and nothing else, which is why the MRTS term is inert against a non-stationary second moment. Where the mean surface is nonlinear the term does improve the score, once the basis is fine enough to resolve that surface, and it converts a small part of the headroom the true surface shows to be available. Where it converts most, it is repairing a misspecified partition rather than adding to a fit that is already well specified. On the ozone data neither extension separates from its matched baseline.

The implication extends beyond ozone, since the same target, a spatial probability statement about the upper tail at unmonitored sites, arises for any heavy-tailed, asymmetric environmental field regulated by a high threshold. Temporal order and mean flexibility are not free improvements to such a model, but neither are they idle. Rather than replacing the STP baseline, the standardized AR(2) prior and the MRTS mean enlarge it into a candidate class from which held-out tail scores can select the setting that fits the data at hand. Where each axis fell short marks where to go next. Entering the MRTS basis with random coefficients, as in fixed-rank kriging, would place the flexibility where the simulation showed the structure to be. A parameterization of z that removes the scale ridge of Section 3.3, together with longer records, would test whether the forecast gains survive at horizons beyond the fifteen days examined here, and whether they appear on observed rather than simulated dynamics.

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A Notation

Several letters carry more than one meaning in this thesis. Table 9 lists those reused symbols together to make the intended reading explicit; symbols not listed here are used with a single meaning.

Table 9: Symbols that reuse a letter or shape, with their distinct meanings.

Symbol	Meaning
$\phi_1,\ \phi_2$	AR(2) coefficients
$\phi^{(w)},\ \phi^{(z)},\ \phi^{(\sigma)}$	AR(1) coefficients of the knot-location, skewness, and scale processes
φ	Matérn spatial range
Φ	Standard normal distribution function
$\psi_i(\mathbf{s}),\ \boldsymbol{\psi}(\mathbf{s}),\ \boldsymbol{\Psi}$	Thin-plate spline kernel, its vector, and $n \times n$ matrix
γ_h	AR stationary autocovariance, constrained so $\gamma_0 = 1$
η	Matérn proportion of spatial variance, $\eta \in [0, 1]$
$\Gamma(\cdot)$	Gamma function (in the Matérn correlation)
λ	Skew-t skewness parameter
$\mu_1 \geq \cdots \geq \mu_n$	MRTS eigenvalues (of $\mathbf{Q}\,\boldsymbol{\Psi}\,\mathbf{Q}$)
$\mathbf{X}_t(\mathbf{s}),\ \mathbf{X}_t^*(\mathbf{s})$	Regression and MRTS-augmented covariate vectors
$\mathbf{R},\ \mathbf{r}$	MRTS low-order polynomial design matrix and a single-site row; $\mathbf{R}(\varphi, \nu, \eta)$ is instead the Matérn correlation matrix of (26)
K	Number of partition knots
K_{MRTS}	Number of MRTS basis functions
K_ν	Modified Bessel function of the second kind, order ν
$\sigma_t(\mathbf{s}),\ \sigma_{tk}^2$	Skew-t scale process

Table 9 (continued)

Symbol	Meaning
σ_ϵ^2	AR innovation variance
$\sigma_\beta^2,\ \sigma_\lambda^2$	Prior variances of $\boldsymbol{\beta}$ and λ
σ_g	Error scale of the nonlinear-mean designs
$Y_t,\ Y_t(\mathbf{s})$	Generic scalar latent AR process in the Yule–Walker and MCMC derivations; $Y_t(\mathbf{s})$ is instead the observed response at site $\mathbf{s}$
$z_{tk},\ z_{tk}^*$	Latent skewness variable and its PIT-transformed value; $\lvert z_t(\mathbf{s})\rvert$ in (1) is the half-normal skew term
h	Spatial distance $\lVert\mathbf{s}_1 - \mathbf{s}_2\rVert$; also the temporal lag in the Yule–Walker context, and the forecast lead in Section 2.6 and the time-block evaluation
H	Number of leads forecast from one origin
$g(\mathbf{s})$	Fixed nonlinear mean surface
$g_j(\mathbf{s})$	j-th cosine field of the random effect
$v_t(\mathbf{s})$	Standard Gaussian (Matérn) process
$\mathbf{v}_k$	k-th eigenvector (column of $\mathbf{V}$)
n	Number of locations that build the MRTS basis
$n_s,\ n_t,\ n_o$	Number of sites; number of days; length of the observed record at a forecast origin
$M,\ m$	Number of retained posterior draws, and its index, in Algorithm 1

B Stationary moments of the standardized AR(2)

We derive the Yule–Walker moments (13)–(14) used to standardize the AR(2) prior (11) to unit marginal variance. Let $\gamma_h = \mathrm{Cov}(Y_t, Y_{t-h})$ be the stationary autocovariances, constrained so that $\gamma_0 = \mathrm{Var}(Y_t) = 1$. Multiplying the recursion (11) by Y_{t-h} for $h \geq 1$ and taking expectations, and using that ϵ_t is independent of Y_{t-h} for the causal stationary solution, yields the Yule–Walker recursion

$$\gamma_h \;=\; \phi_1\,\gamma_{h-1} \;+\; \phi_2\,\gamma_{h-2}, \qquad h \geq 1. \tag{38}$$

Evaluating (38) at $h = 1$ and using the symmetry $\gamma_{-1} = \gamma_1$ with $\gamma_0 = 1$ gives $\gamma_1 = \phi_1 + \phi_2\,\gamma_1$, hence $\gamma_1 = \phi_1/(1 - \phi_2)$, and at $h = 2$ gives $\gamma_2 = \phi_1\,\gamma_1 + \phi_2$; together these are (13). Multiplying (11) instead by Y_t and taking expectations, with $\mathrm{E}[Y_t\,\epsilon_t] = \sigma_\epsilon^2$ because Y_t carries the current innovation, gives $\gamma_0 = \phi_1\,\gamma_1 + \phi_2\,\gamma_2 + \sigma_\epsilon^2$, and solving for σ_ϵ^2 under $\gamma_0 = 1$ recovers (14). The same normalization fixes the initial pair, which has unit variances and lag-1 covariance γ_1 and is therefore the stationary bivariate Gaussian (15).

C AR(2) full conditionals for the Metropolis-within-Gibbs update

Updating the time-varying parameters. The AR(2) Markov structure factorizes the joint prior of $(Y_1, \ldots, Y_{n_t})$ as

$$p(Y_1, \ldots, Y_{n_t}) \;=\; p(Y_1, Y_2) \prod_{s=3}^{n_t} p(Y_s \mid Y_{s-1}, Y_{s-2}), \tag{39}$$

with $p(Y_1, Y_2)$ given by (15) and each conditional factor for $s \geq 3$ following (11). For an interior time point $3 \leq t \leq n_t - 2$, the value Y_t enters (39) through three factors, namely the self factor $p(Y_t \mid Y_{t-1}, Y_{t-2})$, the lag-1 factor $p(Y_{t+1} \mid Y_t, Y_{t-1})$, and the lag-2 factor $p(Y_{t+2} \mid Y_{t+1}, Y_t)$. The full conditional targeted by the Metropolis step for Y_t is therefore

$$\pi(Y_t \mid Y_{-t}, \mathbf{Y}) \;\propto\; L(\mathbf{Y} \mid Y_t) \cdot \underbrace{p(Y_t \mid Y_{t-1}, Y_{t-2})}_{\text{self}} \cdot \underbrace{p(Y_{t+1} \mid Y_t, Y_{t-1})}_{\text{lag-1}} \cdot \underbrace{p(Y_{t+2} \mid Y_{t+1}, Y_t)}_{\text{lag-2}}, \tag{40}$$

where $L(\mathbf{Y} \mid Y_t)$ is the spatial likelihood contributed by Y_t through the partition and Gaussian process structure of (1). The lag-2 factor is absent for $t = n_t - 1$ and both lag-1 and lag-2 factors are absent for $t = n_t$. At the head of the series the initial-pair density (15) plays the role of the self factor. Accordingly, Y_1 enters only the initial-pair factor $p(Y_1, Y_2)$ and the conditional $p(Y_3 \mid Y_2, Y_1)$, while Y_2 enters $p(Y_1, Y_2)$, $p(Y_3 \mid Y_2, Y_1)$, and $p(Y_4 \mid Y_3, Y_2)$; in each case the factors referencing a non-existent time index are simply dropped. Under a symmetric Gaussian random-walk proposal $Y_t^{(c)} \sim N(Y_t, s^2)$, the Metropolis–Hastings acceptance ratio is

$$R \;=\; \min\!\left(1,\; \frac{\pi(Y_t^{(c)} \mid Y_{-t}, \mathbf{Y})}{\pi(Y_t \mid Y_{-t}, \mathbf{Y})}\right), \tag{41}$$

in which all three prior factors must appear; omitting the lag-2 factor distorts the stationary distribution of the chain away from the AR(2) joint and produces a lag-2 autocorrelation that does not match the Yule–Walker value in (13). The proposal scale is adapted during the first half of the burn-in period, by factors 0.8 and 1.2 whenever the running acceptance rate leaves $[0.25, 0.50]$, and is held fixed thereafter, so that the sampling phase runs a time-homogeneous kernel. The move is componentwise for z^* and σ^{2*}, while for the knot locations the $2K$ coordinates of a given day are proposed and accepted as a single block, which leaves (41) unchanged in form because the block proposal is again a symmetric random walk.

Updating the autoregressive coefficients. Restoring the knot index, write Y_{tk} for the transformed series of knot k and $N(x;\, \mu,\, \sigma^2)$ for the normal density evaluated at x. Conditional on those series, $\boldsymbol{\phi}^{(\cdot)}$ is independent of the data and of every other parameter, so its full conditional carries only the AR(2) prior and the coefficient prior of (31),

$$\pi\big(\boldsymbol{\phi}^{(\cdot)} \mid Y\big) \;\propto\; \left[\prod_{k=1}^{K} p\big(Y_{1k}, \ldots, Y_{n_t k} \mid \boldsymbol{\phi}^{(\cdot)}\big)\right] N_2\big(\boldsymbol{\phi}^{(\cdot)};\, \mathbf{0},\, \sigma_\phi^2 \mathbf{I}_2\big)\, I\big(\boldsymbol{\phi}^{(\cdot)} \in \mathcal{S}\big), \tag{42}$$

the product running over the K knots, which are independent replicates of the same recursion, and over both coordinates in the case of $\mathbf{w}^*$. Expanding each factor by (39) and discarding $p(Y_{1k}) = N(0,1)$, which does not involve $\boldsymbol{\phi}^{(\cdot)}$, the log-likelihood in (42) is

$$\ell\big(\boldsymbol{\phi}^{(\cdot)}\big) \;=\; \sum_{k=1}^{K} \log N\big(Y_{2k};\, \gamma_1 Y_{1k},\, 1-\gamma_1^2\big) \;+\; \sum_{k=1}^{K}\sum_{t=3}^{n_t} \log N\big(Y_{tk};\, \phi_1 Y_{t-1,k} + \phi_2 Y_{t-2,k},\, \sigma_\epsilon^2\big), \tag{43}$$

with γ_1 and σ_ϵ^2 the functions of $\boldsymbol{\phi}^{(\cdot)}$ fixed by (13)–(14). The first sum is the contribution of the initial pair and cannot be dropped, because it depends on the coefficients through $\gamma_1 = \phi_1/(1-\phi_2)$. Keeping only the transitions would leave the coefficient update targeting the conditional likelihood while the update of Y_t above targets the full joint, so that the two blocks of the sampler no longer share a stationary distribution. Under AR(1) the same omission is harmless, since there the only discarded factor is $p(Y_1) = N(0,1)$ and the day-2 factor is itself a transition.

The two coefficients are drawn one at a time, the second conditioning on the value of the first just accepted. For ϕ_1 a candidate is proposed from the untruncated random walk $\phi_1^{(c)} \sim N(\phi_1, s_\phi^2)$ and, when $(\phi_1^{(c)}, \phi_2) \in \mathcal{S}$, is accepted with probability

$$\min\left(1,\ \frac{\exp\{\ell(\phi_1^{(c)}, \phi_2)\}\ N\big(\phi_1^{(c)};\, 0,\, \sigma_\phi^2\big)}{\exp\{\ell(\phi_1, \phi_2)\}\ N\big(\phi_1;\, 0,\, \sigma_\phi^2\big)}\right), \tag{44}$$

while a candidate falling outside $\mathcal{S}$ is rejected outright and the chain holds at its current value. The coefficient ϕ_2 is drawn the same way with the roles of the two coefficients exchanged, and s_ϕ is adapted by the rule stated above. Rejecting outside $\mathcal{S}$ needs no Hastings adjustment. The proposal is symmetric and the acceptance region $\mathcal{S}$ does not depend on the current state. For any two points of $\mathcal{S}$ the proposal density is therefore the same in both directions, detailed balance holds with respect to (42) restricted to $\mathcal{S}$, and the mass proposed outside $\mathcal{S}$ only raises the probability of staying put. A proposal truncated to $\mathcal{S}$ together with the matching asymmetry correction, the scheme used by the AR(1) baseline, is equally valid but defines a different chain. One pair $\boldsymbol{\phi}^{(\cdot)}$ is shared by all K knots of its process.

D Marginal covariance of the non-stationary-dependence design

The second MRTS design of Section 3.4 places the non-stationarity in the dependence through a mean-zero random effect built from two fixed cosine fields on the $[0,10]^2$ domain. With centers $\mathbf{c}_1 = (0,10)^\top$, $\mathbf{c}_2 = (7.5, 2.5)^\top$ and wavenumbers $\kappa_1 = \pi/10$, $\kappa_2 = \pi/5$,

$$g_j(\mathbf{s}) = \cos\big(\kappa_j \, \|\mathbf{s} - \mathbf{c}_j\|\big), \qquad j = 1, 2, \tag{45}$$

and the random effect superposes them with independent, daily-redrawn Gaussian weights,

$$u_t(\mathbf{s}) = \sum_{j=1}^{2} \xi_{tj}\, g_j(\mathbf{s}), \qquad \xi_{tj} \overset{iid}{\sim} N(0, \tau_j^2), \quad (\tau_1, \tau_2) = (5, 3), \tag{46}$$

independently across j and across days t. The response adds u_t to the one-knot skew-t model,

$$Y_t(\mathbf{s}) = 10 + \lambda\, \sigma_t(\mathbf{s})\, |z_t(\mathbf{s})| + \sigma_t(\mathbf{s})\, v_t(\mathbf{s}) + u_t(\mathbf{s}). \tag{47}$$

Marginalizing over the weights, $u_t(\mathbf{s}) = \sum_j \xi_{tj}\, g_j(\mathbf{s})$ has zero mean, $\mathrm{E}[u_t(\mathbf{s})] = \sum_j g_j(\mathbf{s})\, \mathrm{E}[\xi_{tj}] = 0$, and, by the independence of ξ_{t1} and ξ_{t2}, the low-rank covariance

$$\mathrm{Cov}\big(u_t(\mathbf{s}), u_t(\mathbf{s}')\big) \;=\; \sum_{j=1}^{2} \tau_j^2\, g_j(\mathbf{s})\, g_j(\mathbf{s}'). \tag{48}$$

The pointwise variance $\mathrm{Var}\big(u_t(\mathbf{s})\big) = \sum_j \tau_j^2\, g_j(\mathbf{s})^2$ depends on location, so the dependence is non-stationary while the marginal mean stays constant at 10. Because the weights are redrawn independently across days, u_t carries no temporal dependence, keeping this design orthogonal to the AR(2) experiment.